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Decentralized Spatial Negotiation via Utility-guided Residual Multi-agent Reinforcement Learning for Trajectory Generation

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Abstract—Traffic in many developing regions operates under weak lane discipline, heterogeneous vehicle interactions, and decentralized negotiation of shared space. These environments challenge conventional lane-based trajectory prediction and motion planning frameworks, which typically rely on structured map priors and predefined leader-follower relationships. This paper presents a utility-guided residual multi-agent reinforcement learning framework for motion modeling in weakly structured non-lane-based traffic.

The proposed approach combines an interpretable utility-based behavioral model with a residual multi-agent reinforcement learning policy. The utility model encodes directional preference, speed incentive, collision risk, and proximity-aware interaction terms derived from behavioral and economic decision-making principles. Rather than learning motion policies from scratch, the reinforcement learning layer adaptively modulates the utility-driven behavior through residual policy learning, enabling context-dependent negotiation while every executed move remains a discrete utility maximizer under bicycle kinematics.

The framework is evaluated on a weak-lane multilane freeway corridor reconstructed from high-resolution aerial trajectories collected in Lebanon. The residual policy is benchmarked against classical continuous-space planners (ORCA, social force, DWA, MPPI), a Frenet-observation MAPPO baseline, and the calibrated utility prior under a shared closed-loop harness with vehicle-scale OBB lookahead inside the utility kernel. On the sparse freeway protocol, the prior and residual policy are collision-free, while MAPPO records 2.1 contacts with the shortest travel time (33.8 s, arrival 0.93) and pairwise planners record 0.3–3.2. Residual MARL raises sparse arrival relative to the prior (0.89 versus 0.86). Under denser spawn, residual collisions fall from 8.5 to 7.1 while MAPPO rises to 44.9 despite 0.97 arrival. Freezing kernel-scale residuals reverses the dense residual gain. Every residual move remains inside the same interpretable utility maximizer.

This work establishes a principled framework for integrating behavioral utility theory and multi-agent reinforcement learning for decentralized motion generation in weakly structured traffic environments. The code and pre-trained models are available at: <https://github.com/pebeigi/U-ResMARL>.

Index Terms—Motion Modeling, Reinforcement Learning, Multi-Agent Systems, Decision-Making Under Uncertainty, Trajectory Data, Model Calibration

I. INTRODUCTION

In many developing countries, transportation systems operate under conditions that differ substantially from the structured, lane-disciplined traffic typically assumed in classical models [1]–[3]. Weak rule enforcement, heterogeneous vehicle

fleets, and informal negotiation of right-of-way produce complex interactions that challenge traditional traffic flow theory [4]. Such irregular behaviors including lateral encroachments, opportunistic yielding, and abrupt accelerations are prevalent in Lebanon, where infrastructure and enforcement limitations contribute to weak-lane and non-standard driving patterns [5]. Accurately representing these settings requires empirical data and modeling approaches that remain valid when lane structure and deterministic leader-follower assumptions are only weakly satisfied.

Recent advances in aerial data collection enable direct observation of naturalistic traffic with high spatial and temporal resolution. Drone-based video provides an unobstructed top-down view of vehicle motion, allowing large-scale extraction of trajectories while preserving behavioral authenticity [6]. This study focuses on an aerial trajectory dataset collected in Jounieh, Lebanon, over a multilane freeway segment characterized by non-lane-based longitudinal interactions and informal lateral negotiation. Roundabout geometries with opportunistic merging remain an important future evaluation setting, but all residual-learning and benchmarking experiments reported here are confined to the freeway corridor so that claims match the calibrated prior and closed-loop harness.

Microscopic vehicle models and modern trajectory forecasting frameworks remain central to representing vehicle interactions. However, state-of-the-art deep learning architectures (e.g., LaneGCN, VectorNet, Wayformer) predominantly assume strict lane centerlines, predefined map topologies, and structured drivable spaces. Consequently, they struggle to capture decentralized negotiation, emergent lane formation, and interaction-only forecasting without map priors in weak-lane environments. Furthermore, while Multi-Agent Reinforcement Learning (MARL) offers a pathway to game-theoretic trajectory generation, standard MARL formulations frequently fail in traffic applications. These failures stem from enormous action spaces, sparse rewards, collision instability, and a lack of human behavioral priors, ultimately yielding physically unrealistic driving behavior.

To bridge this gap, we introduce a utility-guided residual MARL architecture specifically designed for unstructured, non-lane-based traffic. By decoupling human behavioral priors from deep policy learning, the proposed framework avoids

the pitfalls of blind reinforcement learning. The architecture operates through a four-stage process:

- **Stage 1: Utility Prior:** An interpretable baseline utility formulation (U_{base}) captures directional preference, collision risk, and speed incentive. The additive structure is prospect-*inspired* (asymmetric progress/risk trade-offs) rather than a Kahneman–Tversky value/weighting function.
- **Stage 2: Local Neighborhood Encoding:** Each agent observes a fixed budget of nearest neighbors (relative position and velocity) together with corridor clearances, yielding a fully decentralized observation.
- **Stage 3: RL Residual:** Instead of learning raw control actions, a shared residual policy learns a parameter adjustment $\Delta\Theta(o_i)$ that modulates the underlying utility weights and collision-kernel scales.
- **Stage 4: Final Utility Maximization:** The executed action is obtained by discrete utility maximization under $\Theta_i = \text{clip}(\Theta_{base} + \Delta\Theta_i)$, preserving physical realism, interpretability, and kinematic feasibility while enabling context-dependent negotiation.

Beyond reproducing vehicle kinematics, weak-lane and chaotic driving reflect asymmetric human decision-making under perceived risk. Prospect theory provides a principled framework to quantify such behaviors relative to reference points, highlighting how drivers trade off progress against conflict risk. The proposed residual MARL framework complements this by providing a data-driven mechanism to capture implicit interaction rules and spatial negotiation across multiple agents in dense settings.

The contributions of this study are:

- Calibration of a twelve-parameter utility prior (including vehicle-scale collision-kernel widths) on Lebanon freeway aerial trajectories, with robust working parameters for downstream simulation.
- A utility-guided residual MARL policy that learns nine-dimensional parameter adjustments $\Delta\Theta$ (including $\Delta\sigma_{||}, \Delta\sigma_{\perp}$) rather than raw controls, preserving interpretability under bicycle kinematics.
- A matched-seed closed-loop benchmark on the freeway corridor comparing residual MARL to ORCA, social force, DWA, MPPI, MAPPO, and the frozen utility prior, with a kernel-scale ablation and a dense-negotiation stress protocol.
- Evidence that the lookahead-equipped prior is collision-free on the sparse protocol; residual MARL then improves sparse arrival and lowers mean dense collisions relative to the prior (only when $\Delta\sigma$ is enabled), whereas unconstrained MAPPO wins arrival by leaving the human speed band and colliding under crowding.

This integrated approach establishes a robust foundation for modeling heterogeneous, non-lane-based traffic using a principled combination of deterministic, stochastic, and learning-based methods.

II. LITERATURE REVIEW

Modeling vehicle motion in traffic systems has been a central topic in transportation research, with a wide range of approaches developed to describe how drivers interact with their surroundings. Traditional models largely rely on structured, lane-based assumptions, while more recent work has explored interaction-driven, behavioral, and learning-based frameworks to better capture complex and unstructured traffic environments. This section reviews the relevant literature across four key domains: classical microscopic models, non-lane-based traffic modeling, interaction-based motion models, and data-driven learning approaches.

A. Classical Microscopic Traffic Models

Microscopic traffic flow models describe the behavior of individual vehicles, typically focusing on how drivers adjust acceleration in response to surrounding traffic conditions. Among the most widely used formulations are car-following models such as the Intelligent Driver Model (IDM) [7], [8], the Full Velocity Difference Model (FVDM) [9], and Gipps' model. These models assume a well-defined leader–follower relationship and describe vehicle acceleration as a function of spacing, relative speed, and desired velocity.

Such formulations have been successful in modeling traffic dynamics on structured highways with strong lane discipline. Their parameters are often interpretable, capturing driver preferences for desired speed, time headway, and acceleration behavior. However, their reliance on a single leading vehicle and strictly longitudinal interactions limits their applicability in environments where lane structure is weak or absent. In such cases, interactions become inherently two-dimensional, and drivers may respond simultaneously to multiple neighboring vehicles rather than a single leader.

B. Non-Lane-Based and Heterogeneous Traffic

In many developing regions, traffic is characterized by weak lane discipline, heterogeneous vehicle types, and informal negotiation of space. These conditions give rise to non-lane-based traffic, where vehicles do not adhere to fixed lanes and instead move in a continuous two-dimensional space. Several studies have highlighted the limitations of classical car-following models in such environments and proposed adaptations to account for lateral interactions and multiple leaders.

Research on heterogeneous traffic flow emphasizes the importance of capturing interactions beyond a single leader, including lateral encroachments and opportunistic maneuvers. Approaches often rely on data-driven definitions of interaction neighborhoods or extend classical models to consider multiple surrounding vehicles. Despite these efforts, many models retain a fundamentally longitudinal structure and struggle to fully represent the decentralized and multi-agent nature of non-lane-based traffic. This has motivated the exploration of alternative modeling paradigms that operate directly in continuous space.

C. Interaction-Based and Continuous-Space Models

To address the limitations of lane-based formulations, a class of models has emerged that represents vehicle motion through local interactions in continuous space. One prominent example is the Social Force Model (SFM) [10], originally developed for pedestrian dynamics. In this framework, agents are subject to attractive forces toward desired motion and repulsive forces from nearby agents, enabling flexible modeling of dense and unstructured interactions.

In robotics and multi-agent systems, collision avoidance has been extensively studied using velocity-based approaches such as Velocity Obstacles (VO) and Reciprocal Velocity Obstacles (RVO). Building on these concepts, the Optimal Reciprocal Collision Avoidance (ORCA) framework [?], [?] formulates collision avoidance as a set of linear constraints in velocity space. Each agent computes a feasible set of collision-free velocities by considering pairwise interactions with neighboring agents and selects the velocity closest to its preferred motion.

These approaches are well-suited for modeling multi-agent interactions in continuous space and can scale to dense environments without requiring explicit lane definitions. However, they are primarily geometric and deterministic in nature, focusing on collision avoidance rather than behavioral realism. As a result, they do not explicitly capture driver heterogeneity, risk perception, or decision-making under uncertainty, which are critical components of real-world driving behavior.

D. Behavioral and Stochastic Models

Human driving behavior is inherently stochastic and influenced by subjective perceptions of risk and reward. Behavioral models aim to capture these aspects by incorporating psychological and decision-theoretic principles into motion modeling. Prospect Theory [?] provides a well-established framework for modeling decision-making under risk, where outcomes are evaluated relative to a reference point and losses are weighted more heavily than gains.

In transportation research, prospect-theoretic formulations have been applied to problems such as route choice and gap acceptance, demonstrating the importance of asymmetric risk perception in driving decisions. These models provide a natural way to represent aggressive or conservative behavior, particularly in environments where drivers must continuously evaluate trade-offs between progress and safety.

Despite their strengths, most behavioral models are not formulated in a fully continuous two-dimensional setting and often rely on simplified interaction assumptions. Integrating behavioral decision-making with realistic multi-agent interactions remains an open challenge, particularly in non-lane-based traffic environments.

E. Learning-Based and Multi-Agent Approaches

Recent advances in machine learning have led to the development of data-driven approaches for modeling vehicle behavior, particularly through reinforcement learning. In this paradigm, agents learn control policies by interacting with an

environment and optimizing a reward function that encodes desired objectives such as safety, efficiency, and comfort.

Multi-Agent Reinforcement Learning (MARL) extends this framework to settings where multiple agents interact simultaneously. The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm [?] is a widely used approach for continuous control problems, enabling agents to learn decentralized policies while leveraging centralized training. Such methods have been applied to autonomous driving scenarios, demonstrating the ability to capture complex interaction patterns and emergent behaviors.

Learning-based approaches offer significant flexibility and can adapt to complex, high-dimensional environments without requiring explicit modeling assumptions. However, they often lack interpretability and depend heavily on the design of reward functions and training data. Moreover, their behavior may be difficult to analyze or generalize beyond the training conditions.

F. Research Gap and Contribution

The existing literature provides a diverse set of tools for modeling traffic dynamics, ranging from deterministic interaction-based models to stochastic behavioral formulations and data-driven learning approaches. However, these paradigms are typically studied in isolation, and direct comparisons across fundamentally different modeling assumptions remain limited.

In particular, there is a lack of unified frameworks that evaluate deterministic collision-avoidance models, stochastic decision-theoretic models, and multi-agent learning approaches within the same non-lane-based traffic context. This gap is especially relevant for heterogeneous traffic environments, where interactions are multi-dimensional and behavior is highly variable.

To address this limitation, the present study develops a comparative freeway evaluation that places a calibrated utility prior and a residual MARL adapter alongside classical continuous-space planners (ORCA, social force, DWA, MPPI) and unconstrained MAPPO in a shared closed-loop harness. The comparison is designed so that residual MARL is judged as a calibrated, interpretable generator—not as a substitute for pairwise geometric avoidance or a speed-maximizing RL policy.

G. Trajectory Data

The rapid growth of aerial data collection has transformed how chaotic and non-lane-based traffic patterns are observed and modeled. Unmanned aerial vehicles (UAVs) provide a top-down view that can simultaneously capture trajectories of all road users, including vehicles, cyclists, and pedestrians, over complex geometries. A UAV-based methodological framework was developed for multivehicle trajectory extraction at urban roundabouts, demonstrating that drones can capture detailed kinematic profiles and gap-acceptance behavior that are difficult to observe using fixed cameras [11]. Other researchers

used drones to analyze traffic flow at urban roundabouts, showing how high-resolution aerial imagery can reveal subtle lane encroachments, yielding patterns, and circulation strategies that are often invisible in aggregated loop-detector data [12].

In parallel, several open datasets have emerged that provide high-fidelity trajectories from aerial views for intersections and roundabouts. Researchers introduced the TUMDOT-MUC dataset, a large-scale collection of multimodal trajectories (motorized vehicles and vulnerable road users) gathered by drone over multiple urban sites, specifically designed to support model calibration and safety analysis in complex multimodal environments [13]. Other researchers published a dataset of UAV-based aerial images with detailed vehicle annotations, enabling researchers to build and validate computer-vision pipelines for vehicle detection and, by extension, trajectory extraction in circular intersections [14]. More recently, the MiTra dataset was released, which provides drone-based trajectories for freeways with ramps across all traffic states, illustrating how persistent aerial observation can be used to study microscopic behavior from free flow to congestion [15].

Aerial data has also been leveraged for proactive safety analysis. The Extra Brake Required to Avoid a Crash (EBRAC) surrogate measure was proposed using drone-based image-recording data, quantifying near-crash conflicts at intersections by reconstructing detailed braking profiles and interaction dynamics [16]. This work illustrates how high-resolution trajectories can be transformed into interpretable safety indicators that directly reflect road-user maneuvers. Similar methodologies have been applied to other drone-based datasets to derive conflict-based safety metrics and evaluate how geometric design and control strategies affect risk.

H. Research Gap

Overall, the literature shows (i) strong evidence that weak lane discipline and heterogeneous traffic produce irregular, continuous-space interactions; (ii) a split between geometric collision-avoidance planners, behavioral utility models, and unconstrained MARL; and (iii) high-resolution aerial trajectories that make those methods testable on the same naturalistic scenes. What is still missing is a single generator that is *at once* (a) calibrated to observed choice, (b) kinematically constrained and interpretable, and (c) able to adapt that prior when crowding increases—without collapsing into either a pure collision-avoidance oracle or a reward-hacking speed policy.

This paper closes that gap on a Lebanon freeway corridor by calibrating a twelve-parameter utility prior, training a residual MARL adapter over that prior, and comparing the resulting policy in a shared harness to ORCA, social force, DWA, MPPI, MAPPO, and the frozen prior. Residual MARL occupies the intended intersection of those requirements; the two urban datasets in Appendix E probe whether the prior itself transfers when corridor geometry is replaced by destination-plus-curb encoding, without repeating the residual-learning study.

III. METHODOLOGY

This work proposes a utility-guided residual multi-agent reinforcement learning framework for motion generation in weakly structured non-lane-based traffic. Unlike conventional lane-based trajectory prediction approaches that rely on explicit lane markings, discrete lane topology, or predefined leader-follower relationships, the proposed framework models vehicle interactions directly in continuous two-dimensional space through decentralized utility maximization and adaptive behavioral learning.

Although lane markings are not imposed as discrete behavioral modes, agents remain constrained by the physical geometry of the drivable corridor. Corridor-aligned Frenet features and boundary penalties are therefore used to encode physical road geometry without reducing behavior to classical lane keeping.

The proposed framework combines two complementary components:

- 1) an interpretable utility-based behavioral prior that captures directional preference, collision risk, speed regulation, proximity/progress incentive, and corridor adherence;
- 2) a residual multi-agent reinforcement learning policy that adaptively modulates selected utility parameters according to the surrounding traffic context.

Rather than directly learning low-level control actions or trajectories from scratch, the reinforcement learning layer learns residual behavioral adaptations over a structured utility formulation. This preserves physical interpretability and kinematic feasibility while enabling emergent negotiation behaviors in dense heterogeneous traffic.

A. Problem Formulation

Traffic is modeled as a decentralized multi-agent system operating in continuous two-dimensional space. Let $\mathcal{A} = \{1, 2, \dots, N\}$ denote the set of traffic agents. Each agent $i \in \mathcal{A}$ is represented by a continuous kinematic state

$$s_i^t = [x_i^t, y_i^t, v_i^t, \theta_i^t], \quad (1)$$

where (x_i^t, y_i^t) denotes spatial position, $v_i^t \geq 0$ denotes speed magnitude, and θ_i^t denotes heading angle.

Each agent observes a local neighborhood of at most K nearest neighbors within a perception radius R_p :

$$\mathcal{N}_i^t = \left\{ j \in \mathcal{A} \setminus \{i\} \mid \|r_j^t - r_i^t\|_2 \leq R_p \right\}_{1:K}, \quad (2)$$

where $r_i^t = (x_i^t, y_i^t)$ and the subset retains the K closest neighbors by Euclidean distance. In all closed-loop experiments we set $K = 6$ and $R_p = 60$ m.

The objective of each agent is to generate feasible motion that progresses along the corridor, avoids collisions, remains within drivable boundaries, and negotiates shared space with neighboring agents. Unlike classical lane-based formulations, agents are not restricted to discrete lane IDs or fixed leader-follower roles.

B. Utility-Based Behavioral Prior

The nominal motion behavior of each agent is governed by a utility-maximization process over a discrete control-candidate set. At each time step, agent i evaluates candidates generated from a finite grid of longitudinal acceleration and steering values,

$$\mathcal{G} = \{(a, \delta)\} = \mathcal{A}_{\text{acc}} \times \mathcal{A}_{\text{steer}}, \quad (3)$$

and selects the candidate that maximizes expected utility after one-step kinematic bicycle rollout:

$$(a_i^*, \delta_i^*) = \arg \max_{(a, \delta) \in \mathcal{G}} U_i(s'_i(a, \delta); \Theta_i), \quad (4)$$

where $s'_i(a, \delta)$ is the predicted next state obtained from the bicycle dynamics and Θ_i denotes the agent's behavioral utility parameters.

The total utility consists of five components:

$$U_i = U_{\text{dir}} + U_{\text{speed}} + U_{\text{dist}} - U_{\text{col}} - U_{\text{path}}. \quad (5)$$

1) *Directional Utility*: Directional utility encourages motion aligned with the local corridor tangent rather than an unconstrained destination vector. Let

$$\hat{m}_i = \frac{r'_i - r_i}{\|r'_i - r_i\|_2} \quad (6)$$

denote the unit move direction induced by a candidate next position r'_i , and let $\hat{t}_i$ denote the unit tangent of the corridor centerline at the ego agent's current projection. Then

$$U_{\text{dir}} = S_\theta \hat{m}_i^\top \hat{t}_i, \quad (7)$$

where $S_\theta > 0$ controls directional sensitivity. This formulation remains valid for both positive- and negative-direction traffic because directionality is defined by the corridor orientation rather than the global x -axis.

2) *Speed Incentive*: Speed incentive rewards candidate rollout speeds that are close to a desired speed $v_{\text{des},i}$. Because scalar speed is non-negative, reverse motion is represented through heading rather than negative speed. For a candidate speed v'_i , define the symmetric closeness ratio

$$\rho_i = \frac{\min(v'_i, v_{\text{des},i})}{\max(v'_i, v_{\text{des},i})} \in (0, 1]. \quad (8)$$

The speed utility is then

$$U_{\text{speed}} = S_v \frac{\rho_i}{1 + \rho_i^{(\xi_i - 1)/2}}, \quad (9)$$

where $S_v > 0$ scales the term and $\xi_i > 1$ controls how sharply the reward decreases under speed mismatch. The maximum occurs at $v'_i = v_{\text{des},i}$, and overspeeding and underspeeding are penalized symmetrically.

3) *Proximity Incentive*: Proximity utility encourages progress toward a local reference point expressed in corridor Frenet coordinates. Let (s'_i, n'_i) be the longitudinal and lateral projections of the candidate next position onto the corridor, and let $(s_{\text{ref}}, n_{\text{ref}})$ be the corresponding projection of a local reference location (e.g., a short-horizon target state). The anisotropic effective distance is

$$d_{\text{eff}} = w_x |s'_i - s_{\text{ref}}| + w_y |n'_i - n_{\text{ref}}|. \quad (10)$$

A speed-dependent perception horizon,

$$H_p = \max(\kappa v_i, H_{\min}), \quad (11)$$

normalizes this distance, yielding

$$U_{\text{dist}} = \frac{S_d}{1 + (d_{\text{eff}}/H_p)^\gamma}, \quad (12)$$

where $S_d > 0$ caps the reward magnitude and $\gamma > 0$ controls how quickly the incentive decays for distant candidates.

4) *Collision-Risk Utility*: In weakly structured environments, collision risk cannot be reduced to hard geometric overlaps alone. Instead, collision risk is modeled by comparing the candidate next pose of the ego agent with short-horizon predictions of neighbors, expressed in the ego body frame and measured relative to vehicle footprints (length L , width W ; defaults $4.5 \text{ m} \times 1.8 \text{ m}$).

For neighbor j , the predicted future position after the candidate arrival time Δt is obtained by constant-velocity extrapolation,

$$\mu_j = r_j + v_j \Delta t. \quad (13)$$

Let $\Delta r_{i'j} = R(\psi_{i'})^\top (r'_i - \mu_j)$ be this relative vector in the candidate heading frame ($+x$ forward). Surface gaps beyond the same-size footprints are

$$g_{\parallel} = \max(0, |\Delta r_{i'j}^{(x)}| - L), \quad g_{\perp} = \max(0, |\Delta r_{i'j}^{(y)}| - W), \quad (14)$$

so overlapping oriented boxes yield zero gap. Soft presence uses an anisotropic Gaussian on these gaps,

$$P_{j,i'} = \exp\left(-\frac{1}{2} \left[\left(\frac{g_{\parallel}}{\sigma_{\parallel}}\right)^2 + \left(\frac{g_{\perp}}{\sigma_{\perp}}\right)^2 \right]\right), \quad (15)$$

with calibrated (and residual-adapted) scales $\sigma_{\parallel}, \sigma_{\perp}$. Aggregating over neighbors and capping cumulative risk yields

$$U_{\text{col}} = w_c \min\left(\sum_{j \in \mathcal{N}_i} P_{j,i'}, 1\right), \quad (16)$$

where $w_c > 0$ controls collision aversion. Candidate selection additionally rejects bicycle rollouts whose predicted OBB overlaps a neighbor when a collision-free alternative exists, and speed alignment uses the nearest ahead neighbor as a car-following reference when one is present.

Calibration scores this kernel at the data/control interval Δt (one predicted neighbor pose). Closed-loop residual training and benchmarking additionally query the same OBB kernel along a short constant-velocity horizon (1.5 s, four substeps).

Geometric planners in the benchmark do *not* use that lookahead at decision time; they are scored with the same OBB contact detector after each executed step.

5) *Path Adherence and Boundary Constraints*: Although agents negotiate laterally inside the corridor without discrete lane constraints, they remain subject to infrastructural boundaries. A path-adherence penalty discourages proximity to corridor edges and off-road intrusion:

$$U_{\text{path}} = w_\ell (1 - e^{-\beta \ell_i^2}), \quad (17)$$

where ℓ_i is a non-negative lateral clearance residual relative to the corridor boundary (including an optional near-edge buffer), w_ℓ scales the penalty magnitude, and β controls the sharpness of the transition near the boundary.

C. Kinematic Bicycle Candidate Generation

Each control candidate (a, δ) is rolled out through a one-step kinematic bicycle model. Given the current ego state (x, y, v, θ) and wheelbase L ,

$$\begin{aligned} v' &= \text{clip}(v + a\Delta t, 0, v_{\max}), \\ \theta' &= \theta + \frac{v}{L} \tan \delta \Delta t, \\ x' &= x + v' \cos \theta' \Delta t, \\ y' &= y + v' \sin \theta' \Delta t. \end{aligned} \quad (18)$$

Thus the utility evaluation is performed over the resulting candidate next state (x', y', v', θ') , rather than over free speed/heading pairs. Importantly, v' is constrained to be non-negative; negative-direction travel is represented by heading aligned with the corridor orientation.

D. Residual Multi-Agent Reinforcement Learning

Fixed utility parameters cannot fully capture adaptive negotiation, contextual yielding, temporary aggressiveness, or emergent interaction strategies. Therefore, a residual multi-agent reinforcement learning (MARL) policy modulates a subset of the utility parameters according to the local observation.

The residual policy is

$$\Delta \Theta_i^t = \pi_\phi(o_i^t), \quad (19)$$

and the adapted parameters are

$$\Theta_i^t = \text{clip}(\Theta_i^{\text{base}} + \Delta \Theta_i^t). \quad (20)$$

Motion is then selected by utility maximization under the adapted parameters:

$$(a_i^*, \delta_i^*) = \arg \max_{(a, \delta) \in \mathcal{G}} U_i(s'_i(a, \delta); \Theta_i^t). \quad (21)$$

Thus the RL policy does not directly output accelerations or trajectories; it only modulates the interpretable behavioral prior.

E. Policy Observation Space

Each agent receives a fully decentralized local observation in corridor Frenet coordinates for the ego vehicle and in the ego body frame for neighbors (world (x, y) on this diagonal corridor saturates a fixed Cartesian scale):

$$o_i^t = \left[s_i, n_i, v_i, \tilde{\theta}_i, \tilde{\theta}_{\text{goal},i}, d_{\text{lower},i}, d_{\text{upper},i}, \{\Delta x_{ij}^{\text{body}}, \Delta y_{ij}^{\text{body}}, \Delta v_{x,ij}^{\text{body}}, \Delta v_{y,ij}^{\text{body}}\}_{j \in \mathcal{N}_i} \right], \quad (22)$$

where (s_i, n_i) are the longitudinal/lateral projections onto the PCA tube, $\tilde{\theta}_i$ and $\tilde{\theta}_{\text{goal},i}$ are heading errors relative to the local tangent, $d_{\text{lower},i}$ and $d_{\text{upper},i}$ are clearances to the corridor bounds, and missing neighbors are zero-padded up to the fixed neighbor budget K .

F. Residual Action Space

The policy adapts both the magnitude and sensitivity of selected utility components, including the collision-kernel scales. The continuous residual action is nine-dimensional:

$$\Delta \Theta_i^t = [\Delta S_v, \Delta S_\theta, \Delta S_d, \Delta w_c, \Delta \xi_i, \Delta \gamma, \Delta w_\ell, \Delta \sigma_{\parallel}, \Delta \sigma_{\perp}]. \quad (23)$$

The remaining utility parameters (w_x, w_y, β) are treated as fixed geometric/shape priors in the current implementation. Residual actions are bounded (tanh-scaled per coordinate) and clipped to keep Θ_i^t within physically valid ranges.

G. Reward Function

The residual policy is trained with a multi-objective reward balancing progress, safety, control smoothness, and boundary feasibility:

$$R_i = w_1 R_{\text{progress}} + w_2 R_{\text{safety}} + w_3 R_{\text{smooth}} + R_{\text{boundary}}, \quad (w_1, w_2, w_3) \quad (24)$$

The individual terms are

$$R_{\text{progress}} = v_i \cos(\theta_i - \theta_{\text{tan},i}), \quad (25)$$

where $\theta_{\text{tan},i}$ is the local corridor tangent angle. Let $g_{ij} = \|r_j - r_i\|_2 - L$ be the centre-to-centre gap minus vehicle length. Contact-aware safety is

$$R_{\text{safety}} = \sum_{j \in \mathcal{N}_i} \begin{cases} -16(1 - g_{ij}), & g_{ij} \leq 0, \\ -\exp(-g_{ij}/2), & g_{ij} > 0. \end{cases} \quad (26)$$

An additional penalty of 8 is subtracted from each agent involved in an OBB contact at that step. Smoothness is

$$R_{\text{smooth}} = -(a_i^2 + \lambda_\delta \delta_i^2). \quad (27)$$

$$R_{\text{boundary}} = \begin{cases} -(C_{\text{out}} + c_{\text{depth}} \min(d_{\text{out}}, d_{\text{max}})), & \text{outside corridor,} \\ 0, & \text{otherwise.} \end{cases} \quad (28)$$

where a_i and δ_i are the executed bicycle controls, $\lambda_\delta = 0.5$ weights steering cost in R_{smooth} , $C_{\text{out}} = 20$ is the base off-corridor penalty (exceeding the largest per-step progress

reward at $v_{\max} = 16 \text{ m/s}$, $c_{\text{depth}} = 3$ provides a capped inward gradient, and $d_{\max} = 5 \text{ m}$ prevents return-scale blow-up for untrained policies. An additional per-step collision penalty of 8 is subtracted from every agent involved in an OBB contact. An optional trajectory-matching term may be included when ground-truth imitation targets are available; in the present experiments it is set to zero ($w_{\text{traj}} = 0$). Table VIII lists every reward and boundary constant together with the simulator timing parameters used in training and benchmarking.

H. Training Procedure

The residual policy is trained with *parameter-shared* proximal policy optimization (PPO) on decentralized local observations. One actor-critic network is shared by all agents: the actor maps o_i^t to $\Delta\Theta_i^t$ and the critic estimates $V(o_i^t)$ from the *same* local features. Unlike the MAPPO baseline evaluated in Section IX, the critic does not access a joint multi-agent state during training; the setup is therefore decentralized actor-critic PPO (homogeneous weight sharing), not centralized-training-decentralized-execution (CTDE) in the MAPPO/MADDPG sense. Multi-agent rollouts pool every agent's transitions into a single PPO update; execution at test time uses only o_i^t .

Episodes are initialized from simulated traffic states with spawn rules matched to the Lebanon corridor distribution. At each step:

- 1) each agent receives its local observation o_i^t ;
- 2) the policy outputs residuals $\Delta\Theta_i^t$;
- 3) utility parameters are updated and clipped;
- 4) discrete acceleration/steering candidates are generated and scored by the modulated utility;
- 5) the maximum-utility bicycle transition is executed.

Importantly, the learned policy never directly commands unconstrained trajectories. Vehicle motion remains governed by utility maximization and kinematic bicycle constraints, preserving interpretability while enabling complex emergent interactions in weakly structured traffic. Actor/critic architecture, PPO hyperparameters, $\Delta\Theta$ ranges, Θ clipping bounds, observation normalization, and all simulator constants are tabulated in Appendix D.

IV. DATA COLLECTION AND TRAJECTORY EXTRACTION

This study uses high-resolution aerial vehicle trajectories collected over multilane freeway segments in Jounieh, Lebanon, under weak lane discipline and heterogeneous traffic conditions. Traffic was recorded with a drone in a near-stationary hover, providing a top-down view that reduces occlusions and supports simultaneous tracking of interacting vehicles. The site exhibits pronounced non-lane-based longitudinal and lateral interactions, including informal gap negotiation and shockwave-like disturbance propagation.

Trajectory extraction and site characterization are documented in detail by Khoueiry et al. [17], readers are referred to [17] for the full data-collection protocol and extraction pipeline. In this paper, we restrict attention to passenger-vehicle trajectories with valid kinematic transitions and use

corridor-aligned geometry derived from the observed streams as the spatial reference for utility evaluation. Two additional environments—a compact roundabout in Jounieh and the TGSIM Foggy Bottom street network—are used only for environment-based utility calibration and are documented in Appendix E.

V. CALIBRATION METHODOLOGY

The utility model in Eq. (5) (directional alignment, speed alignment, proximity reward, collision penalty, and boundary adherence) contains twelve free parameters

$$\theta = (S_\theta, S_v, \xi_i, S_d, \gamma, w_x, w_y, w_c, w_\ell, \beta, \sigma_\parallel, \sigma_\perp),$$

including the collision-kernel scales. These parameters are calibrated from the Lebanon naturalistic trajectory set so that the discrete utility prior reproduces observed short-horizon motion before any residual policy is trained.

a) Data and candidate actions.: Valid passenger-vehicle transitions with time step $\Delta t \in [0.05, 0.2] \text{ s}$ yield 190,767 usable state records. From this pool we draw $N_s = 2000$ one-step choice samples. At each sample, the ego agent is reconstructed in the corridor frame with neighboring vehicles inside a 60 m radius (at most six neighbors). Candidate controls are generated on the same acceleration-steering grid used in simulation (acceleration $\in \{-3, \dots, 3\} \text{ m/s}^2$, steering $\in [-0.45, 0.45] \text{ rad}$), producing a discrete action set $\mathcal{A}$. The observed next state is matched to the nearest candidate (target index) for likelihood evaluation.

b) Closed-loop windows.: Because one-step ranking alone under-constrains θ , we split vehicles into train/validation/test (221/74/73) and sample short trajectory windows of horizon $H = 8$ steps (300/100/100). From the observed initial state of each window, the utility model repeatedly selects $\arg \max_{a \in \mathcal{A}} U(a; \theta)$ and the simulated pose/speed/heading are compared with the recorded trajectory. Working parameters are chosen on validation windows; test windows are held out.

c) Objective.: Calibration minimizes a weighted composite loss

$$J(\theta) = \lambda_{\text{cl}} L_{\text{cl}}(\theta) + \lambda_{\text{trk}} \frac{\bar{r}(\theta)}{r_0} + \lambda_{\text{nll}} L_{\text{nll}}(\theta), \quad (29)$$

where L_{cl} is the mean closed-loop tracking error (position, with speed and heading penalties), $\bar{r}$ is the mean rank of the observed-like candidate among $\mathcal{A}$, $r_0 = 63$ normalizes rank, and L_{nll} is the mean softmax negative log-likelihood of the target candidate at temperature $T = 0.25$. We use $\lambda_{\text{cl}} = 1$, $\lambda_{\text{trk}} = 1$, and $\lambda_{\text{nll}} = 0.1$, so closed-loop fidelity is the primary target and the one-step likelihood acts as a light regularizer.

d) Search and robust aggregation.: Because $J(\theta)$ is non-convex and partially flat, we do not trust a single optimizer trajectory. Parameters are drawn uniformly in the search box of Table I. For each of $R = 3$ independent restarts (different random seeds), $N_t = 400$ vectors are screened by the cheap one-step terms; the best 80 candidates per restart (240 scored trials in total) are then evaluated with the closed-loop windows.

TABLE I
SEARCH BOUNDS USED IN UTILITY CALIBRATION.

Parameter	Low	High
S_θ, S_v, S_d	0.05	10
ξ_i	1.1	10
γ, β	0.1 / 0.01	10
w_x, w_y	0.1	10
w_c, w_ℓ	0.01 / 0.1	1000
$\sigma_{\parallel}$ (m)	0.5	5
$\sigma_{\perp}$ (m)	0.3	2.5

Let θ^* be the minimizer of J and define the near-optimal cloud as all scored trials with

$$J(\theta) \leq J(\theta^*) + \max(0.05, 0.05 |J(\theta^*)|).$$

If fewer than three trials satisfy this cutoff, the top ten scored trials are used instead. The *robust* working point θ_{rob} is the coordinate-wise median of that cloud. This aggregation reduces sensitivity to isolated lucky minima while retaining the same predictive performance up to a small objective gap. Quantiles of the top-cloud supply the Sobol box in the appendix (independent sampling of the twelve coordinates; identifiability correlations are not encoded in that design).

VI. CALIBRATION RESULTS

a) Fit quality.: Across the 2000 choice samples, the selected utility prior places the observed-like candidate at mean rank 5.72 (robust working point 6.61). The train-set minimizer attains $J(\theta^*) = 1.317$ (closed-loop loss 0.614), while the robust median θ_{rob} —selected on held-out validation windows—yields $J = 1.184$ (closed-loop loss 0.555 on train windows, 0.576 on validation) and a test closed-loop loss of 0.565. Rank CDFs, margin histograms, and closed-loop loss distributions are summarized in Fig. 1.

b) Working parameters and uncertainty.: Table II reports the single best vector, the robust working point, and top-cloud percentiles after vehicle-scale collision-kernel recalibration. We recommend θ_{rob} for all downstream simulation and residual learning:

$$\begin{aligned} S_\theta &= 4.13, S_v = 2.51, \xi_i = 4.04, S_d = 9.43, \gamma = 0.56, \\ w_x &= 4.85, w_y = 5.83, w_c = 300, w_\ell = 77.6, \beta = 2.14, \\ \sigma_{\parallel} &= 0.89 \text{ m}, \sigma_{\perp} = 0.59 \text{ m}. \end{aligned}$$

Relative to θ^* , the robust set raises the weakly identified collision weight (w_c : 15 $\rightarrow$ 300) and reduces an extreme path weight (w_ℓ : 161 $\rightarrow$ 78), while keeping metre-scale kernel widths consistent with passenger-car half-extents. Fig. 2 shows the global scored-trial distributions (histogram and KDE restricted to the central 95% mass), and Fig. 3 reports the corresponding per-vehicle local-fit distributions.

c) Identifiability.: Of 240 closed-loop-scored trials, only a small subset lies near $J(\theta^*)$; the near-optimal/top cloud still exhibits large coefficient-of-variation across most coordinates (Fig. 4). The strongest near-optimal tradeoff is between S_v and w_c (correlation ≈ 1), with additional coupling among (γ, w_x) ,

$(w_c, \sigma_{\perp})$, and $(S_v, \sigma_{\perp})$. Thus the data identify a *region* of acceptable utility weights more reliably than a unique point estimate.

d) Implications.: For the remainder of this paper we (i) fix θ_{rob} as the nominal utility prior, (ii) treat the reported percentile ranges as the calibrated uncertainty region, and (iii) interpret residual policy learning as refining behavior within an already plausible, but non-unique, utility basin. The appendix Sobol study samples this percentile box independently (it does not reproduce the S_v – w_c correlation).

VII. EXPERIMENTAL SETUP FOR RESIDUAL LEARNING AND BENCHMARKING

All residual-learning and baseline comparisons use a shared closed-loop harness on the Lebanon freeway corridor (run_id= 2, lane Kalman filter branch 1). Agents are integrated with the kinematic bicycle model, collisions are detected with oriented bounding boxes (OBBs), and corridor feasibility is enforced in Frenet coordinates. Scenario seeds, spawn windows, and episode lengths are matched across models so that differences reflect controllers rather than initialization.

a) Residual training.: The utility prior is frozen at the robust calibrated vector $\Theta_{\text{base}} = \theta_{\text{rob}}$ (Table II). A shared-policy PPO residual agent outputs the nine-dimensional $\Delta\Theta$ defined above (including collision-kernel scales $(\Delta\sigma_{\parallel}, \Delta\sigma_{\perp})$). Observations are fully decentralized Frenet/body-frame features. Rewards follow Eqs. (25)–(28), with contact-aware safety, a per-contact OBB penalty, and a depth-capped off-road term. Closed-loop utility evaluation uses the 1.5 s OBB lookahead described above. Unless noted, residual checkpoints are trained under the sparse protocol below.

b) Benchmark models.: The paper-facing comparison set is: ORCA, social force, dynamic window approach (DWA), model-predictive path-integral control (MPPI), MAPPO (pure MARL without a utility prior, retrained on the same Frenet/body-frame observations), the calibrated utility prior, and residual MARL. A Frenet lattice planner and alternative MARL variants (IPPO, HAPPO, HATRPO) remain negative controls (poor arrival/safety trade-offs or unstable returns).

c) Metrics.: We report mean collision events per episode, off-road rate, arrival rate, travel time, RMS jerk, and a realism score defined as the mean 1-D Wasserstein distance between simulated and measured marginals of speed, longitudinal acceleration, and lateral offset (lower is more human-like). Minimum TTC and related safety auxiliaries are logged but not emphasized in the main table.

d) Protocols.: The paper’s target use case is dense spatial negotiation with human-like kinematics, not sparse geometric collision avoidance. *Sparse*: 10 scenarios $\times$ 10 agents with the default spawn window. *Ablation*: the same seeds comparing the prior, residual MARL, and a σ -frozen residual (kernel scales clamped to Θ_{base}). *Dense stress*: 18 agents with a tighter spawn window on the same corridor.

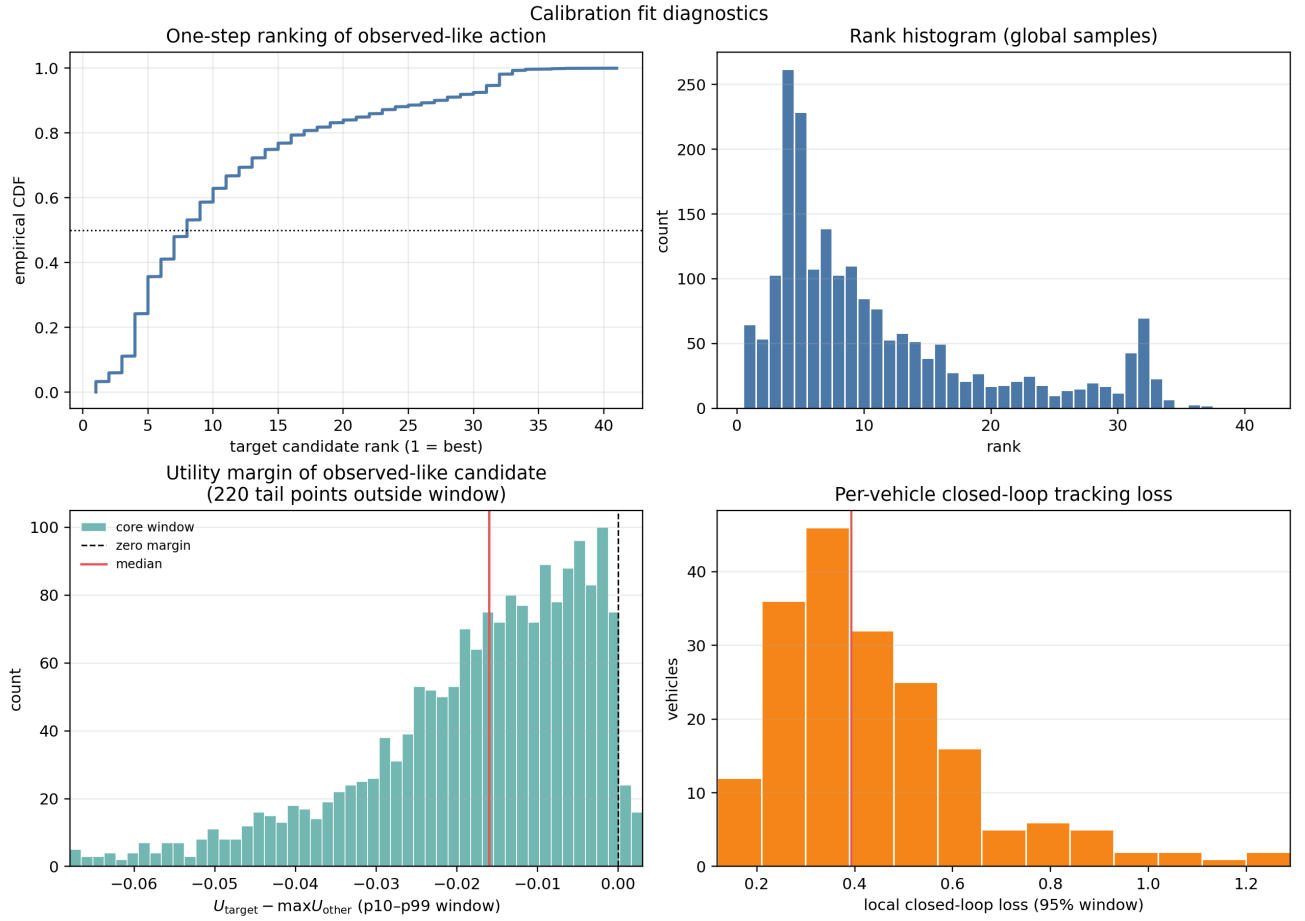


Fig. 1. Calibration fit diagnostics: empirical CDF and histogram of the observed-like candidate rank, utility margin relative to the best alternative (core p10–p99 window), and per-vehicle closed-loop tracking loss.

TABLE II
CALIBRATED UTILITY PARAMETERS AFTER VEHICLE-SCALE COLLISION-KERNEL RECALIBRATION: BEST TRIAL, ROBUST MEDIAN OF THE NEAR-OPTIMAL CLOUD, AND TOP-CLOUD p_{50}/p_{95} .

Param.	Best	Robust	Top p_{50}	Top p_{95}
S_θ	6.744	4.126	5.860	9.147
S_v	0.989	2.512	2.666	8.522
ξ_i	2.318	4.036	4.785	8.989
S_d	9.908	9.432	6.181	9.723
γ	0.388	0.563	3.164	8.211
w_x	3.079	4.853	6.556	9.798
w_y	9.593	5.833	4.214	9.503
w_c	15.12	299.7	255.0	678.9
w_ℓ	161.5	77.64	36.42	135.3
β	2.142	2.142	4.488	8.812
$\sigma_{ }$ (m)	0.889	0.889	2.310	4.574
$\sigma_{\perp}$ (m)	1.338	0.593	1.385	2.020

VIII. RESIDUAL MARL RESULTS

Residual MARL is evaluated first against the frozen utility prior it is designed to improve, then against unconstrained MAPPO, and then against a kernel-scale ablation that tests the contribution of $(\Delta\sigma_{||}, \Delta\sigma_{\perp})$.

Under the sparse protocol (Table III), vehicle-scale OBB lookahead saturates pairwise safety for the entire utility family:

the prior, residual MARL, and the σ -frozen residual all record 0.0 mean collisions. Residual MARL is not a sparse collision-avoidance win over the prior; it is a throughput win inside that collision-free band, raising arrival from 0.86 to 0.89 with comparable travel time. Freezing kernel-scale residuals returns arrival to the prior’s 0.86 and slightly increases RMS jerk (5.35 versus 5.03). MAPPO is the sparse speed leader (travel

Global scored-trial parameter distributions with KDE and 95% intervals (n=240 scored trials, 3 restarts)

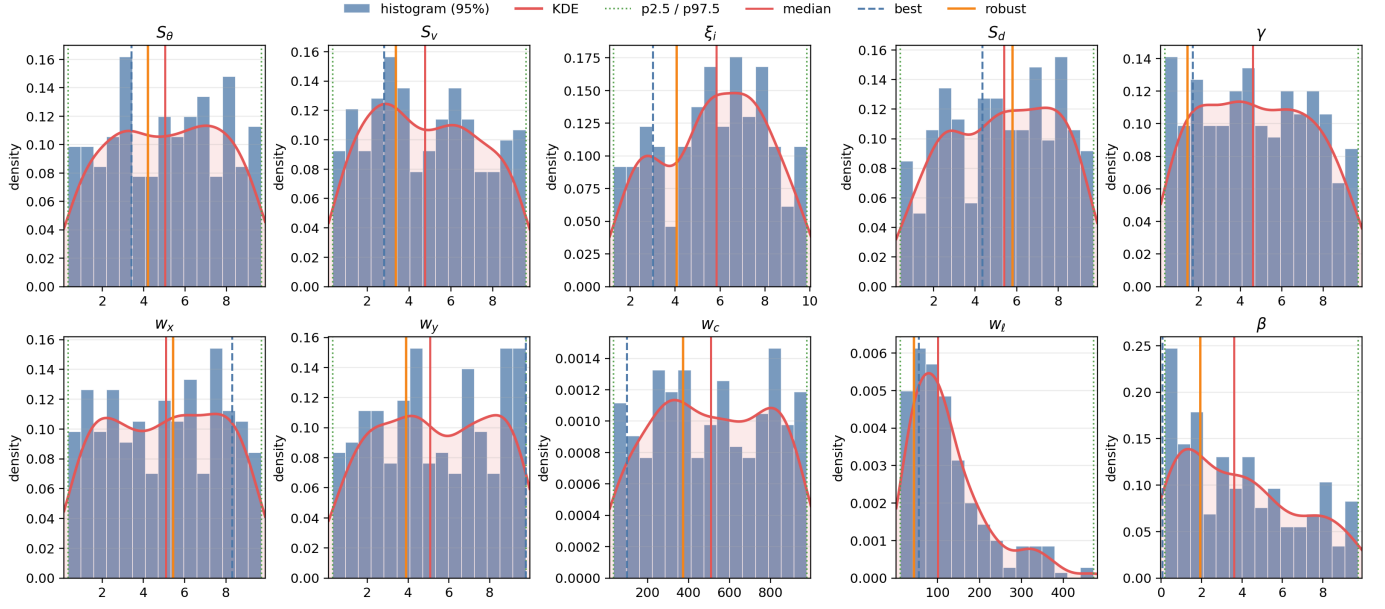


Fig. 2. Global scored-trial distributions of the twelve utility parameters (240 closed-loop-scored trials from three restarts). Histograms and KDEs are computed inside each parameter's central 95% interval; vertical markers indicate the median, best, and robust values when they fall in that window.

Per-vehicle local-fit parameter distributions with KDE and 95% intervals (n=200)

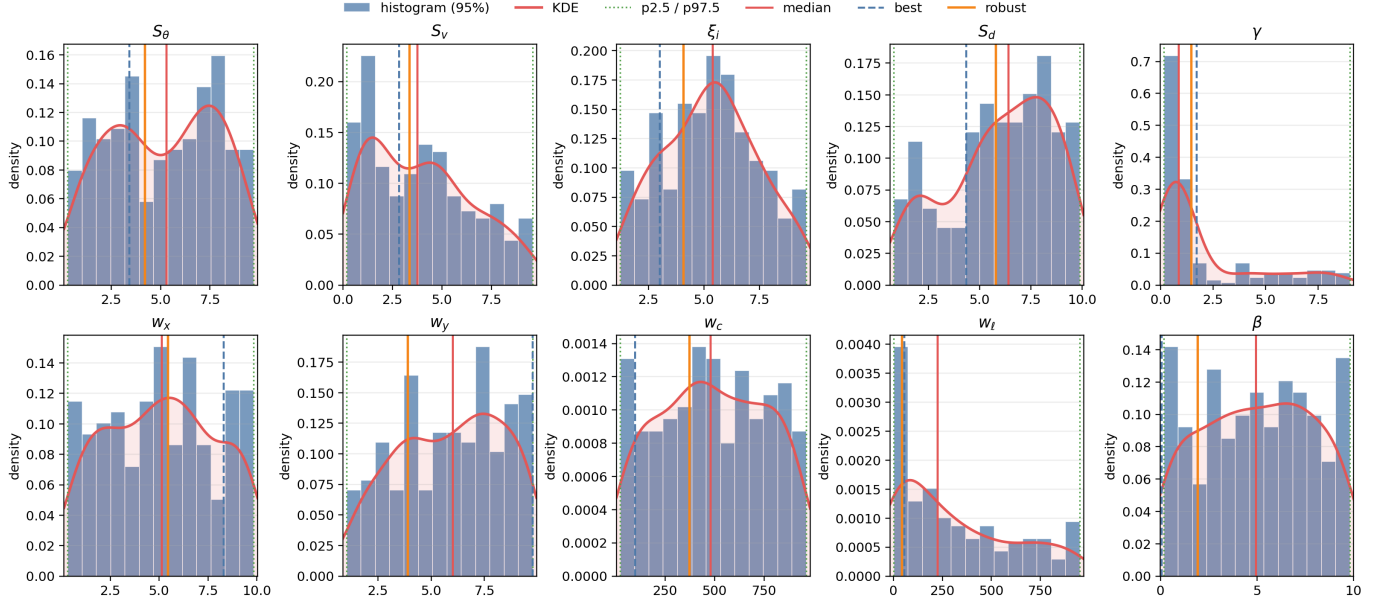


Fig. 3. Per-vehicle local-fit parameter distributions (200 trajectories). Histograms and KDEs use the central 95% mass for each coordinate.

time 33.8 s, arrival 0.93) but records 2.1 collisions and a worse realism score (0.940 versus residual 0.822). Social force edges MAPPO on sparse arrival (0.94) while still colliding (0.3).

The dense protocol (Table IV) is the remaining discriminator. Mean utility-family collisions fall from 8.5 (prior) to 7.1 (residual; $\Delta = -1.4$), while the σ -frozen residual sits at 8.9. MAPPO's sparse arrival advantage transfers as speed, not

safety: collisions rise to 44.9 with 0.97 arrival and 15.1 m/s mean speed. The residual collision Δ versus the prior is not statistically significant at $n = 10$ ($p \approx 0.44$), whereas residual RMS jerk is lower (6.16 versus 6.76). Residual off-road rate increases (0.32 versus 0.24). Kernel adaptation still helps contacts under crowding relative to the prior and is far safer than unconstrained MAPPO, but it is not a free lunch on

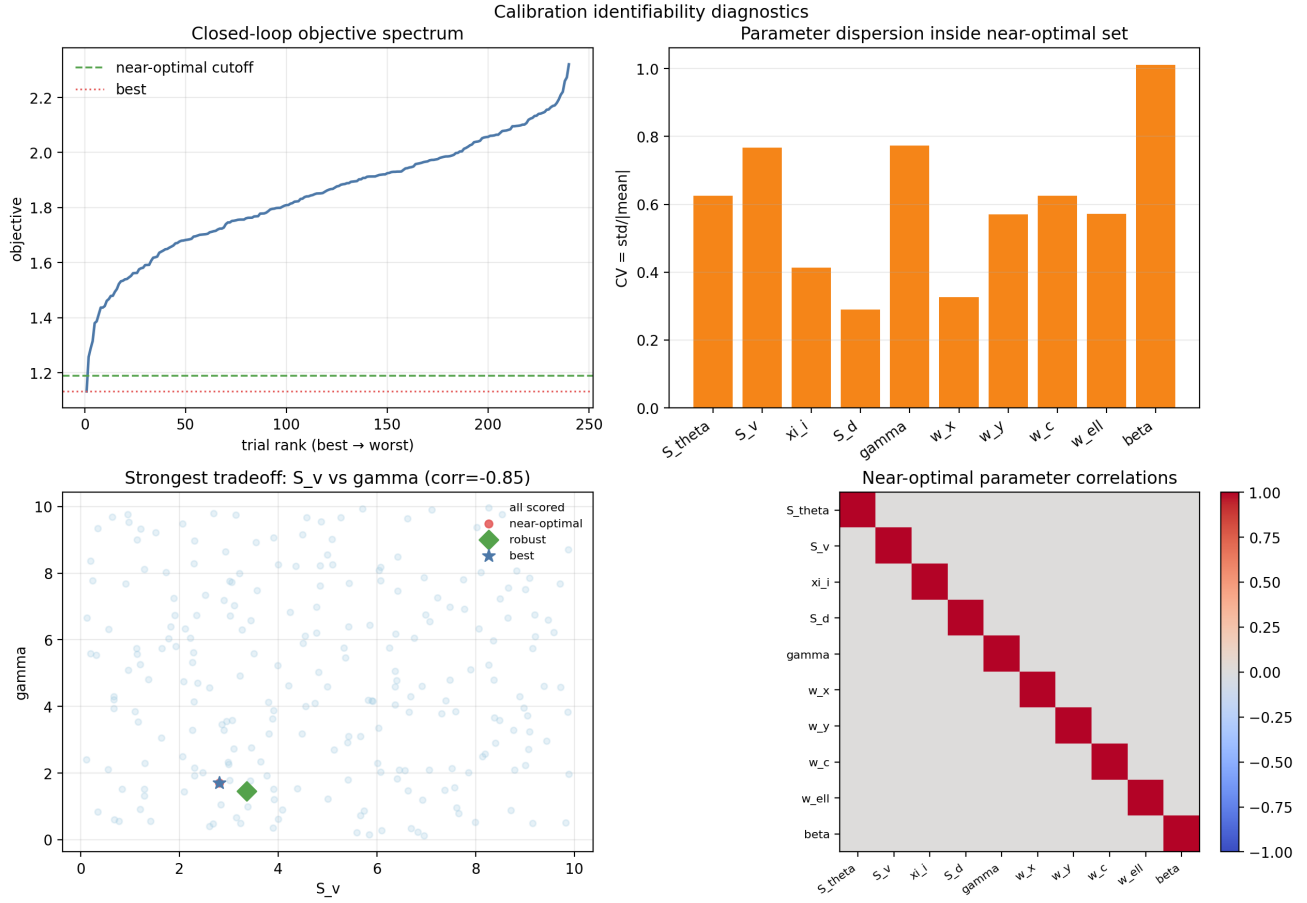


Fig. 4. Identifiability diagnostics for the twelve-parameter freeway fit: pairwise tradeoffs in the near-optimal cloud (notably S_v – w_c) and coefficient-of-variation across coordinates.

TABLE III

SPARSE MATCHED-SEED ABLATION (10 SCENARIOS $\times$ 10 AGENTS). COLLISIONS ARE MEAN EVENTS PER EPISODE. LOOKAHEAD SATURATES SAFETY FOR THE UTILITY FAMILY; RESIDUAL MARL RAISES ARRIVAL RELATIVE TO THE PRIOR, AND KERNEL FREEZING REMOVES THAT THROUGHPUT GAIN.

Model	Collisions	Arrival	Travel (s)	Jerk
Utility prior	0.0	0.86	59.3	4.96
Residual MARL	0.0	0.89	60.2	5.03
Residual (σ frozen)	0.0	0.86	59.6	5.35
MAPPO	2.1	0.93	33.8	0.63

TABLE IV

DENSE STRESS PROTOCOL (18 AGENTS, TIGHTER SPAWN; MEAN OVER 10 SCENARIOS). RESIDUAL MARL LOWERS MEAN COLLISIONS RELATIVE TO THE PRIOR; UNCONSTRAINED MAPPO KEEPS ARRIVAL BY COLLIDING.

Model	Collisions	Arrival	Travel (s)	Jerk
Utility prior	8.5	0.90	55.9	6.76
Residual MARL	7.1	0.91	56.6	6.16
Residual (σ frozen)	8.9	0.92	56.4	6.57
MAPPO	44.9	0.97	34.6	0.64

corridor compliance.

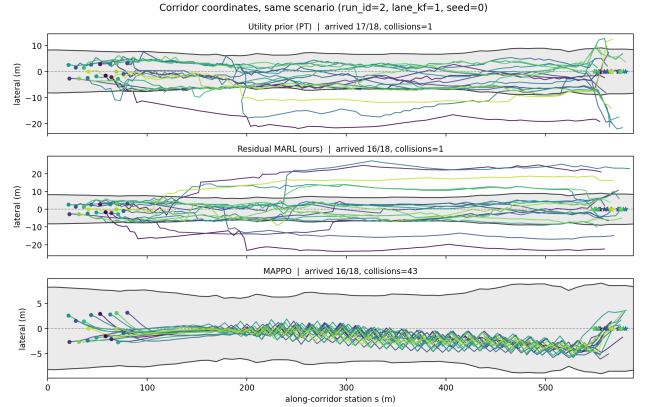


Fig. 5. Dense 18-agent Frenet packing on a matched seed. Residual MARL stays near the prior’s contact count; unconstrained MAPPO packs by colliding.

IX. BENCHMARK

Table V reports the seven-model sparse protocol. With 1.5 s vehicle-scale OBB lookahead *inside the utility kernel*, the calibrated prior and residual MARL are collision-free. MAPPO records 2.1 contacts with the shortest travel time (33.8s;

arrival 0.93). Social force has a slightly higher sparse arrival (0.94) while still colliding (0.3). Pairwise geometric planners still record residual contacts: social force 0.3, ORCA/DWA 0.5, and MPPI 3.2. That comparison is intentional rather than a claim that planners cannot avoid collisions: they optimize their native geometric costs, not the lookahead-equipped utility. Residual MARL has the highest arrival among collision-free controllers (0.89 versus the prior’s 0.86).

Among models that *are* intended as behavioral generators, residual MARL slightly improves the prior’s realism score (0.822 versus 0.836; lower is closer to measured marginals) with comparable travel time. MAPPO’s realism score (0.940) is worse, consistent with a high-speed policy (mean 14.8 m/s versus about 8.3–8.5 for the utility family). The same Wasserstein column does *not* rank residual above DWA/MPPI (0.614–0.640): those planners are smoother (RMS jerk ≈ 0.8 –3.7 versus ≈ 5 for the utility family) and can match speed/acceleration/lateral *marginals* without matching observed discrete choice. We therefore treat realism as a check that residual stays in the prior’s distributional band, not as a claim that the generator is more human-like than a geometric oracle.

Fig. 6 summarizes the sparse metric panel. Fig. 7 shows simulated versus measured speed, acceleration, and lateral-offset distributions. Fig. 5 illustrates packing on the dense protocol, whose numeric ranking is reported in Table IV. Combined, the sparse and dense comparisons support three statements:

- residual MARL improves a calibrated human prior without leaving the interpretable utility maximizer;
- unconstrained MAPPO wins travel time by speeding, then loses safety under crowding (44.9 dense collisions versus residual 7.1);
- geometric planners remain geometric: they optimize avoidance, not observed choice.

X. CONCLUSION

We presented a utility-guided residual MARL framework for decentralized motion on a weak-lane freeway corridor. After calibrating a twelve-parameter utility prior with vehicle-scale collision kernels, a shared residual policy learns nine-dimensional parameter adjustments—including kernel scales—rather than raw controls. On the sparse freeway protocol, OBB lookahead already makes the calibrated prior collision-free; residual MARL then raises arrival from 0.86 to 0.89 without adding contacts. MAPPO is faster (arrival 0.93, 33.8 s) but collides (2.1). Under denser spawn, residual mean collisions are 7.1 versus the prior’s 8.5 and MAPPO’s 44.9; freezing $\Delta\sigma$ returns 8.9. These results support residual utility adaptation as a practical route to context-dependent negotiation without abandoning interpretable behavioral structure.

XI. LIMITATIONS

In calibration, although the calibrated utility prior provides a usable working point for residual learning, several limitations should be noted. First, the twelve-dimensional

utility parameter vector is only weakly identified: multiple substantially different parameter combinations attain nearly the same closed-loop objective, with strong tradeoffs such as that between S_v and w_c . Consequently, the calibration identifies a region of acceptable behavior rather than a unique optimum, which is why we report a robust median of near-optimal trials together with percentile ranges instead of relying on a single best vector. Second, one-step ranking remains imperfect on the discrete candidate grid—the observed-like action is frequently competitive (mean rank about six) but is rarely selected as the unique maximizer—so the one-step likelihood alone under-constrains the parameters and must be complemented by short-horizon closed-loop tracking. Third, per-vehicle local fits exhibit wide dispersion, indicating residual heterogeneity that a single population prior cannot fully capture. Finally, the appendix Sobol study samples the top-cloud percentiles independently, so it does not encode those identifiability correlations; first-order indices in that design are consistent with zero and the total-order ranking is interaction-dominated with wide confidence intervals.

a) Residual learning and benchmarking.: Closed-loop residual and baseline evaluations are freeway-only; residual transfer to the roundabout and Foggy Bottom priors is left for future work. Alternative MARL algorithms (IPPO, HAPPO, HATRPO) exhibited reward hacking or unstable returns in this harness and are not competitive paper baselines. MAPPO was retrained on the current Frenet observations (80 PPO updates, collision penalty 8); it is a single training seed. Dense collision-penalty residual training was not completed for this draft. The dense residual collision improvement (8.5 to 7.1) is modest and not significant at $n = 10$ scenarios, and residual off-road rate is higher than the prior under crowding.

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TABLE V

MAIN SPARSE DIAGNOSTIC ON THE LEBANON FREEWAY CORRIDOR (10 SCENARIOS $\times$ 10 AGENTS; MEAN $\pm$ STD ACROSS SCENARIOS). REALISM IS MEAN WASSERSTEIN DISTANCE TO MEASURED SPEED/ACCELERATION/LATERAL MARGINALS (LOWER IS BETTER). MAPPO IS UNCONSTRAINED MARL; RESIDUAL MARL IS THE PROPOSED BEHAVIORAL GENERATOR.

Model	Coll.	Off-road	Arrival	Travel (s)	Jerk	Realism $\downarrow$
ORCA	0.50 ± 0.92	0.002 ± 0.005	0.850 ± 0.112	63.8 ± 3.9	0.79 ± 0.10	0.720 ± 0.046
Social force	0.30 ± 0.64	0.000 ± 0.000	0.940 ± 0.049	66.8 ± 3.2	0.77 ± 0.07	0.704 ± 0.031
DWA	0.50 ± 0.92	0.000 ± 0.000	0.850 ± 0.102	60.0 ± 1.4	3.66 ± 0.15	0.640 ± 0.028
MPPI	3.20 ± 2.44	0.000 ± 0.000	0.580 ± 0.204	60.5 ± 3.3	1.50 ± 0.21	0.614 ± 0.090
MAPPO	2.10 ± 2.84	0.000 ± 0.000	0.930 ± 0.078	33.8 ± 0.5	0.63 ± 0.07	0.940 ± 0.038
Utility prior	0.00 ± 0.00	0.007 ± 0.011	0.860 ± 0.080	59.3 ± 3.3	4.96 ± 1.32	0.836 ± 0.163
Residual MARL (ours)	0.00 ± 0.00	0.007 ± 0.007	0.890 ± 0.083	60.2 ± 3.7	5.03 ± 1.12	0.822 ± 0.124

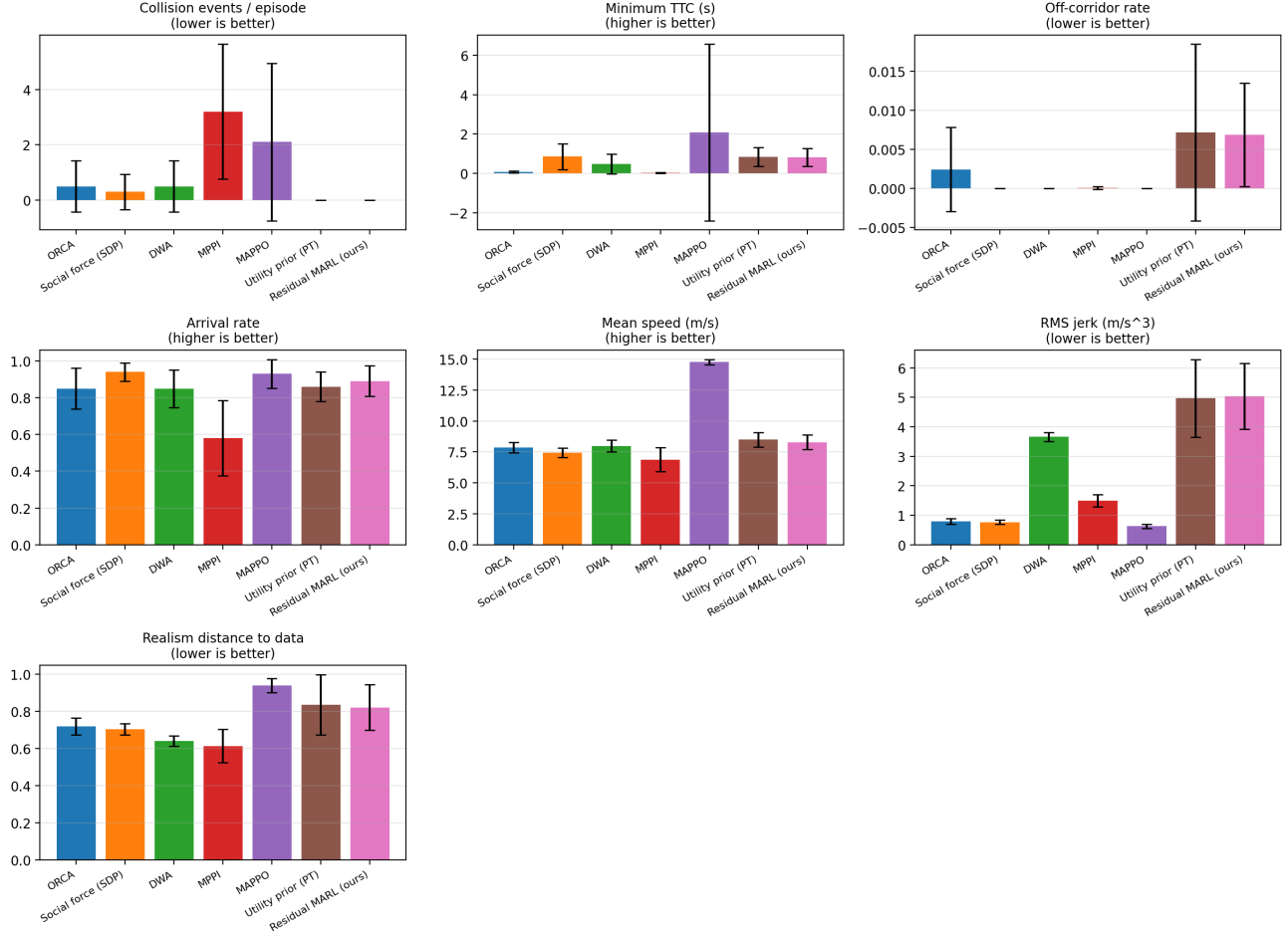


Fig. 6. Sparse protocol metric panel including MAPPO. With OBB lookahead the calibrated prior and residual MARL are collision-free; MAPPO is the speed outlier.

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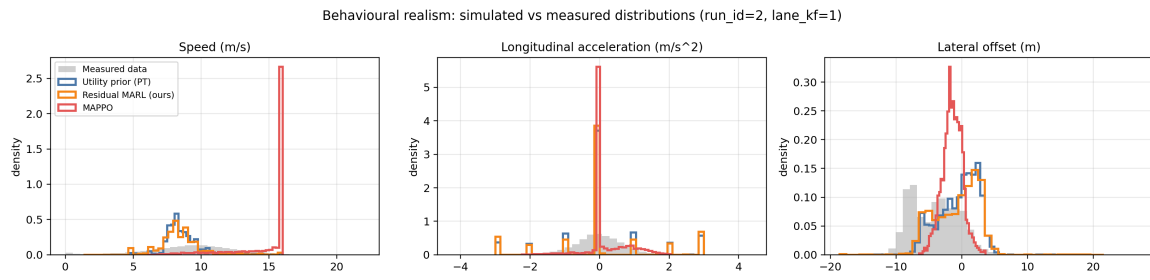


Fig. 7. Realism panel: measured versus simulated speed, acceleration, and lateral-offset marginals. Residual MARL stays in the prior’s distributional band; MAPPO’s speed mass sits well above the measured mode.

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APPENDIX A SENSITIVITY ANALYSIS

Understanding parameter sensitivity is crucial to ensure that the agent-based model produces reliable, realistic, and robust behavior under a wide range of conditions [?], [?]. The sensitivity analysis reveals how robust the model is to parameter variations and helps guide parameter tuning.

To systematically assess model robustness, we perform both local and global sensitivity analyses. Local analysis examines the effect of small variations in individual parameters on each component of the utility function, providing insight into immediate system responsiveness. Global analysis, using techniques such as Sobol indices, quantifies the contribution of each parameter—including interactions—across their entire plausible range. Together, these analyses reveal how sensitive the model is to parameter changes and guide informed tuning for reliable and realistic agent behavior.

APPENDIX B LOCAL SENSITIVITY ANALYSIS

Local sensitivity analysis quantifies how small parameter perturbations affect each component of the utility. We analyze the five utility terms from Equation 5—directional alignment, speed incentive, proximity incentive, collision penalty, and path adherence—separately.

Directional Alignment

Directional utility rewards candidate motion aligned with the local corridor tangent rather than a destination heading. Let $\hat{m}$ denote the unit move direction induced by a candidate next position and $\hat{t}$ the local corridor unit tangent. Then

$$U_{\text{dir}} = S_{\theta} (\hat{m} \cdot \hat{t}). \quad (30)$$

$$\frac{\partial U_{\text{dir}}}{\partial S_{\theta}} = \hat{m} \cdot \hat{t}. \quad (31)$$

$$\frac{\partial U_{\text{dir}}}{\partial (\hat{m} \cdot \hat{t})} = S_{\theta}. \quad (32)$$

Thus S_{θ} linearly scales the directional reward. Utility is maximized when $\hat{m} \cdot \hat{t} = 1$ (perfect corridor alignment) and becomes negative when the candidate move opposes the corridor orientation. This remains valid for both positive- and negative-direction traffic because directionality is defined by corridor geometry.

Speed Incentive

Speed utility rewards candidate bicycle-rollout speeds close to a desired speed. Define the symmetric closeness ratio

$$\rho = \frac{\min(v_{\text{cand}}, v_{\text{des}})}{\max(v_{\text{cand}}, v_{\text{des}})} \in (0, 1], \quad (33)$$

and

$$U_{\text{speed}} = S_v \frac{\rho}{1 + \rho^{(\xi_i - 1)/2}}. \quad (34)$$

Directional Utility Response Surfaces (Corridor Tangent Alignment)

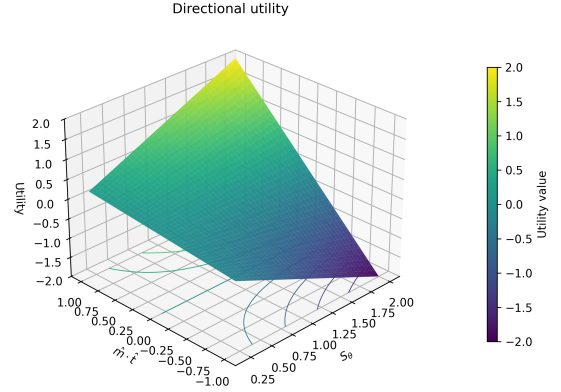


Fig. 8. Directional utility U_{dir} versus corridor alignment $\hat{m} \cdot \hat{t}$ and directional sensitivity S_{θ} .

The corresponding partial derivatives are

$$\frac{\partial U_{\text{speed}}}{\partial S_v} = \frac{\rho}{1 + \rho^{(\xi_i - 1)/2}}, \quad (35)$$

$$\frac{\partial U_{\text{speed}}}{\partial \xi_i} = -\frac{S_v \rho^{(\xi_i + 1)/2} \ln \rho}{2(1 + \rho^{(\xi_i - 1)/2})^2}. \quad (36)$$

Utility attains its maximum at $v_{\text{cand}} = v_{\text{des}}$ ($\rho = 1$). Overspeeding and underspeeding are penalized symmetrically. Larger ξ_i sharpens the drop in utility away from the desired speed, making the agent more sensitive to speed mismatch.

Proximity Incentive

Proximity utility encourages progress toward a local corridor reference expressed in Frenet coordinates. The effective distance is

$$d_{\text{eff}} = w_x |\Delta s| + w_y |\Delta n|, \quad (37)$$

and

$$U_{\text{dist}} = \frac{S_d}{1 + (d_{\text{eff}}/H_p)^\gamma}, \quad H_p = \max(\kappa v_i, H_{\min}). \quad (38)$$

The primary sensitivities are

$$\frac{\partial U_{\text{dist}}}{\partial S_d} = \frac{1}{1 + (d_{\text{eff}}/H_p)^\gamma}, \quad (39)$$

$$\frac{\partial U_{\text{dist}}}{\partial \gamma} = -S_d \frac{(d_{\text{eff}}/H_p)^\gamma \ln(d_{\text{eff}}/H_p)}{(1 + (d_{\text{eff}}/H_p)^\gamma)^2}, \quad (40)$$

$$\frac{\partial U_{\text{dist}}}{\partial d_{\text{eff}}} = -S_d \frac{\gamma (d_{\text{eff}}/H_p)^\gamma}{d_{\text{eff}} (1 + (d_{\text{eff}}/H_p)^\gamma)^2}, \quad (41)$$

$$\frac{\partial U_{\text{dist}}}{\partial H_p} = S_d \frac{\gamma (d_{\text{eff}}/H_p)^\gamma}{H_p (1 + (d_{\text{eff}}/H_p)^\gamma)^2}. \quad (42)$$

Here S_d scales importance, γ controls how quickly the reward decays outside the horizon, w_x and w_y weight longitudinal versus lateral Frenet error, and H_p softens normalized distance as speed increases.

Speed Utility Response Surfaces (Candidate vs Desired)

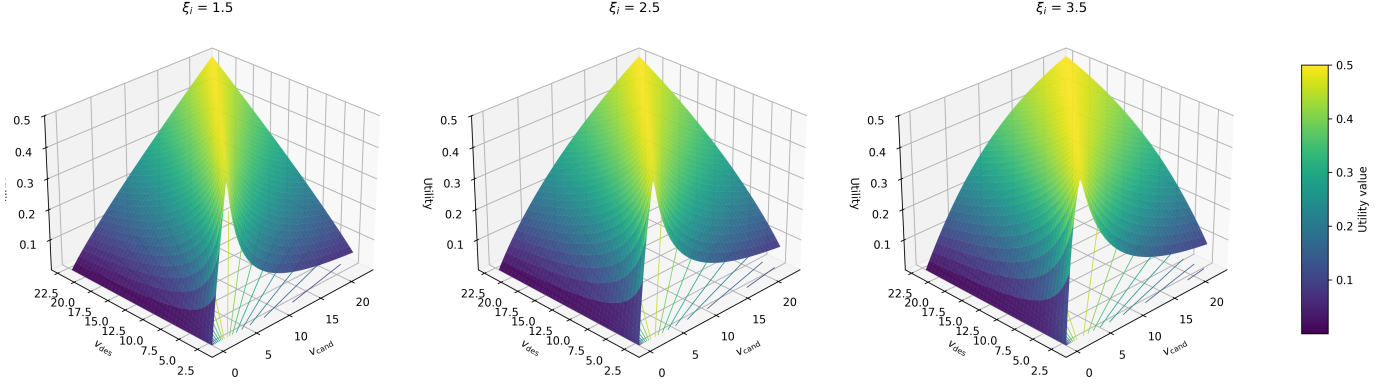


Fig. 9. Speed utility U_{speed} versus candidate speed v_{cand} and desired speed v_{des} (with ξ_i slices).

Proximity Utility Response Surfaces (Frenet Effective Distance)

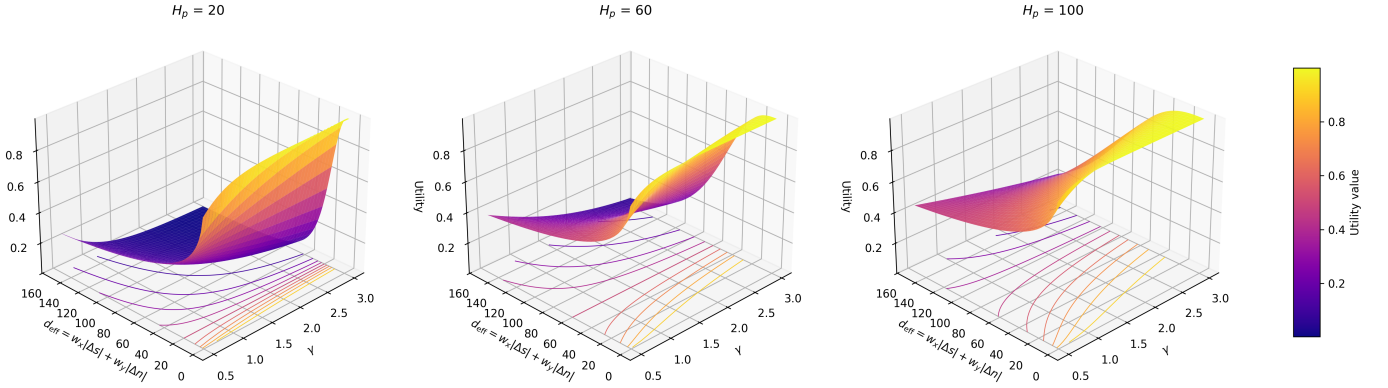


Fig. 10. Proximity utility U_{dist} versus interaction exponent γ , effective Frenet distance d_{eff} , and perception horizon H_p .

Collision-Risk Penalty

The primary partial derivatives for the collision-risk penalty in Eq. (5) are

$$U_{\text{col}} = w_c \min\left(\sum_j P_{j,i'}, 1\right), \quad (43)$$

$$\frac{\partial U_{\text{col}}}{\partial w_c} = \min\left(\sum_j P_{j,i'}, 1\right), \quad (44)$$

with $P_{j,i'} = \exp(-\frac{1}{2}[(g_{\parallel}/\sigma_{\parallel})^2 + (g_{\perp}/\sigma_{\perp})^2])$ on OBB surface gaps. Increasing w_c makes the utility more sensitive to predicted overlap and therefore promotes more risk-averse behavior. Small $(\sigma_{\parallel}, \sigma_{\perp})$ sharpen the risk field around the footprints; large scales broaden it.

Fig. 11 shows this kernel at θ_{rob} : unit presence on overlapping $L \times W$ boxes, then anisotropic decay on the surface gaps $(g_{\parallel}, g_{\perp})$.

Path Adherence

The path-adherence term and its derivatives are

$$U_{\text{path}} = w_{\ell}(1 - e^{-\beta \ell_i^2}), \quad (45)$$

$$\frac{\partial U_{\text{path}}}{\partial w_{\ell}} = 1 - e^{-\beta \ell_i^2}, \quad (46)$$

$$\frac{\partial U_{\text{path}}}{\partial \beta} = w_{\ell} \ell_i^2 e^{-\beta \ell_i^2}, \quad (47)$$

$$\frac{\partial U_{\text{path}}}{\partial \ell_i} = 2 w_{\ell} \beta \ell_i e^{-\beta \ell_i^2}. \quad (48)$$

Here ℓ_i is a non-negative corridor-boundary residual, w_{ℓ} scales the penalty magnitude, and β controls how quickly the penalty saturates near the boundary. Larger w_{ℓ} enforces stricter corridor adherence but may conflict with collision avoidance or progress when set excessively high.

APPENDIX C GLOBAL SENSITIVITY ANALYSIS

Global Sensitivity Analysis (GSA) attributes variance in a simulation-level output Y to uncertainty in the utility parameters, including interactions. This study uses the twelve-parameter top-cloud p_{05} – p_{95} box in Table VI, which contains

Collision Utility Response Surfaces (OBB surface-gap kernel at θ_{rob})

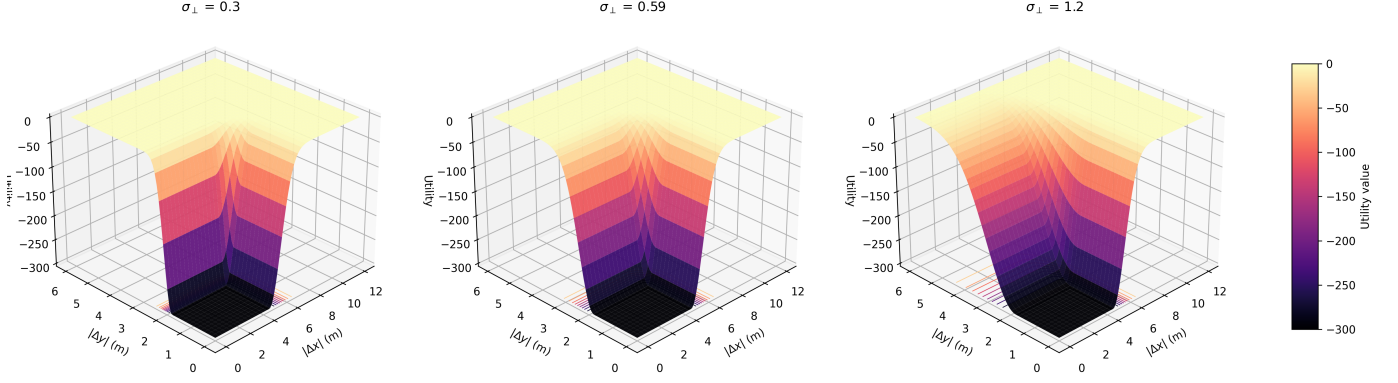


Fig. 11. Collision utility $U_{\text{col}} = -w_c P$ versus body-frame centre separations $|\Delta x|, |\Delta y|$ at $w_c = 300$, $\sigma_{\parallel} = 0.89$ m, $L \times W = 4.5 \times 1.8$ m. Panels vary $\sigma_{\perp}$; the flat well is the overlapping-footprint plateau.

Path-Adherence Utility Response Surfaces

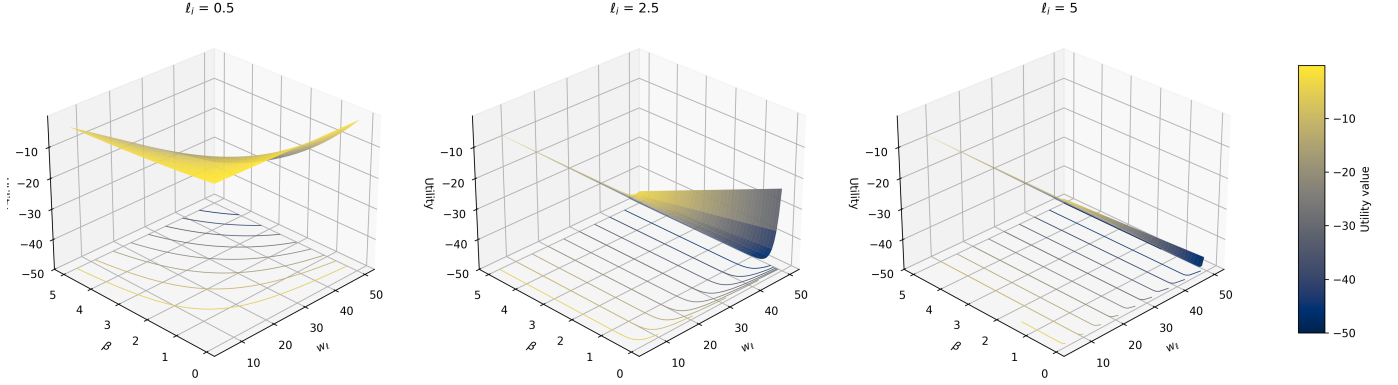


Fig. 12. Path-adherence utility U_{path} versus path-deviation coefficient β and lateral deviation ℓ_i .

θ_{rob} . Samples are drawn independently; they do not reproduce identifiability correlations such as $S_v - w_c$. Each evaluation runs a three-agent follow/merge scene for up to 80 steps ($\Delta t = 0.5$ s) with the kinematic-bicycle candidate grid, the OBB surface-gap collision kernel, and OBB overlap counting.

The scalar output is

$$Y = \sum_{k \in \mathcal{A}} d_k^{\text{final}} + N_{\text{coll}} C_{\text{penalty}}, \quad (49)$$

where d_k^{final} is leftover destination distance if agent k has not arrived, N_{coll} counts pair-timesteps with overlapping oriented boxes, and $C_{\text{penalty}} = 10$. Lower Y is better. Across the 14336 evaluations, Y is heavy-tailed (mean 15.7, std. 18.9, max 407).

The sensitivity analysis is performed using the variance-based Sobol method [?], [?]. Let the model output be

$$Y = f(X_1, X_2, \dots, X_D),$$

where X_j represents independent model parameters. The output variance is decomposed as

$$\text{Var}(Y) = \sum_{j=1}^D V_j + \sum_{1 \leq j < k \leq D} V_{jk} + \dots + V_{12\dots D}. \quad (50)$$

The first-order Sobol index is

$$S_j = \frac{V_j}{\text{Var}(Y)}, \quad (51)$$

while the total-order Sobol index is

$$S_{T_j} = 1 - \frac{\text{Var}_{\mathbf{X}_{\sim j}}(\mathbb{E}_{X_j}[Y | \mathbf{X}_{\sim j}])}{\text{Var}(Y)}. \quad (52)$$

The first-order index quantifies the direct contribution of parameter X_j to the output variance, whereas the total-order index measures its overall contribution, including all interaction effects.

Sampling uses Sobol sequences in SALib with $N = 1024$ and first- plus total-order estimators,

$$N_{\text{eval}} = N(D + 2) = 1024(12 + 2) = 14336.$$

Table VI lists the input box.

TABLE VI
SOBOL INPUT BOX: FREEWAY TOP-CLOUD p_{05} – p_{95} (CONTAINS θ_{rob}).

Parameter	Description	Range
S_θ	Directional weight	[1.39, 9.15]
S_v	Speed-incentive weight	[0.17, 8.52]
ξ_i	Speed-mismatch steepness	[1.49, 8.99]
S_d	Proximity-incentive weight	[3.46, 9.72]
γ	Proximity decay steepness	[0.40, 8.21]
w_x	Longitudinal Frenet weight	[3.00, 9.80]
w_y	Lateral Frenet weight	[1.68, 9.50]
w_c	Collision-risk penalty weight	[20.9, 679]
w_ℓ	Path-adherence penalty weight	[3.33, 135]
β	Path-adherence saturation rate	[0.25, 8.81]
$\sigma_{\parallel}$ (m)	Longitudinal kernel scale	[0.62, 4.57]
$\sigma_{\perp}$ (m)	Lateral kernel scale	[0.58, 2.02]

A. Sensitivity Analysis Results

Table VII reports first-order and total-order indices. Small negative S_1 values are Monte Carlo noise around zero.

TABLE VII
SOBOL INDICES ON THE CALIBRATED TWELVE-PARAMETER BOX
($N = 1024$).

Parameter	S_1	$S_{1,\text{conf}}$	S_T	$S_{T,\text{conf}}$
S_θ	0.008	0.050	0.804	0.246
S_v	−0.018	0.051	0.776	0.267
ξ_i	−0.040	0.067	0.963	0.688
S_d	−0.022	0.034	0.340	0.204
γ	−0.015	0.061	0.838	0.479
w_x	−0.046	0.052	0.798	0.480
w_y	0.006	0.055	0.661	0.311
w_c	−0.020	0.054	0.864	0.316
w_ℓ	−0.042	0.074	1.335	0.635
β	0.017	0.056	1.067	0.813
$\sigma_{\parallel}$	0.036	0.065	1.042	0.453
$\sigma_{\perp}$	0.002	0.068	1.119	0.361

Every first-order index is consistent with zero (S_1 lies within its bootstrap interval of 0). No coordinate, varied alone, explains a stable share of $\text{Var}(Y)$ inside this box. Total-order indices are large for almost all coordinates ($S_T \in [0.34, 1.34]$), so the variance is interaction-dominated. That is the expected Sobol signature when identifiability correlations are ignored: independently combining, for example, a large S_v with a large w_c produces off-manifold priors whose overlap counts swing Y .

Collision-kernel coordinates are no longer silent. w_c , $\sigma_{\parallel}$, and $\sigma_{\perp}$ have total-order mass 0.86, 1.04, and 1.12, unlike the previous ad-hoc box where $w_c \in [0.1, 20]$ never reached the operating value and $S_T(w_c) = 0$. Path terms w_ℓ and β also have large S_T . Proximity weight S_d is the weakest total-order coordinate (0.34). Several $S_T > 1$ estimates, and the wide confidence intervals, reflect sampling error on a heavy-tailed Y rather than a precise ranking.

We therefore do not replace Fig. 4 with a unique Sobol order. The GSA supports two statements about the calibrated box: main effects are small, and collision-kernel plus path-adherence coordinates participate strongly through interactions.



Fig. 13. Aerial frame of the Jounieh study area (weak-lane, heterogeneous traffic) [17].

APPENDIX D REPRODUCIBILITY AND TRAINING CONFIGURATION

Table VIII–IX collect every constant needed to reproduce the closed-loop residual trainer (RL/train_ppo.py) and the shared benchmark harness (Baselines/). Values are identical for evaluation unless noted. The MAPPO baseline uses a separate centralized critic (joint corridor features concatenated with o_i); its PPO settings are listed in the last block of Table IX.

APPENDIX E DATASETS

The main text evaluates residual MARL on a single weak-lane freeway corridor. The utility prior itself, however, is geometry-aware rather than lane-ID-based, so the same five-term family (5) can be instantiated on qualitatively different roadways if the directional, proximity, and path terms are evaluated in a site-appropriate spatial frame. This appendix documents three naturalistic environments used for that purpose (Fig. 16): the Lebanon Highway freeway that supports the main residual-learning experiments, a compact multi-arm roundabout also collected in Jounieh, and the TGSIM Foggy Bottom urban street network in Washington, DC. Residual MARL, planner comparatives, and the dense-stress protocol are *not* repeated on the two urban sites; those recordings are used only to test whether a calibrated utility prior remains meaningful when the corridor assumption is replaced by a destination-plus-curb encoding.

Table X summarizes the calibration-ready samples. All three sources are subsampled to $\Delta t \approx 0.1$ s. For the two urban sites we drop any track that is shorter than 10 s, whose 85th-percentile speed is below 1 m/s, or whose travelled path is shorter than 8 m, so that parked and fragmentary objects never enter the ego pool or the neighbor set. The freeway calibration uses passenger vehicles (class 1) with valid kinematic transitions, as in Section V. Urban calibrations use the dominant moving class at each site (class 2 at the roundabout; class 3 passenger cars at Foggy Bottom). Neighbor footprints default to $4.5 \text{ m} \times 1.8 \text{ m}$ unless a site file supplies length/width.

A. Lebanon Highway

The primary study site is a multilane urban freeway in Jounieh, Lebanon, recorded by a near-stationary drone under weak lane discipline (Fig. 15). Extraction and operational details are given by Khoueiry et al. [17]. Two complementary recordings ($\text{run_id} \in \{1, 2\}$) cover the same curved corridor in opposite observational conditions; each recording

TABLE VIII
REWARD, BOUNDARY, AND CLOSED-LOOP SIMULATOR CONSTANTS (FREEWAY PROTOCOL, $\text{run_id}=2$, $\text{lane_kf}=1$).

Reward / boundary	Value	Simulator timing / geometry	Value
w_1 (progress)	1.0	Step Δt	0.5 s
w_2 (safety)	0.5	Episode horizon $T_{\max}$	240 steps (120 s)
w_3 (smooth)	0.2	Agents per episode N	10 (sparse) / 18 (stress)
w_{traj}	0	Spawn s -range	[20, 120] m
λ_δ	0.5	Min. initial spacing	8.0 m
C_{out}	20	Spawn lateral fraction	0.35 of half-width
c_{depth}	3	Destination margin	uniform in [5, 40] m from end
$d_{\max}$	5.0 m	Arrival tolerance	1.0 m along s
OBB collision penalty	8 / agent / step	Vehicle $L \times W$	4.5×1.8 m
K (neighbor budget)	6	Wheelbase	2.8 m
R_p (perception radius)	60 m	$v_{\max}$	16 m/s
κ (prox. horizon)	2.0	$a_{\max}$	4.0 m/s^2
$H_{\min}$	5.0 m	$\delta_{\max}$	0.45 rad
Candidate grid		OBB lookahead	
Accel. (m/s^2)	$\{-3, \dots, 3\}$	Conflict horizon	1.5 s
Steering (rad)	9 values in $[-0.45, 0.45]$	Conflict substeps	4

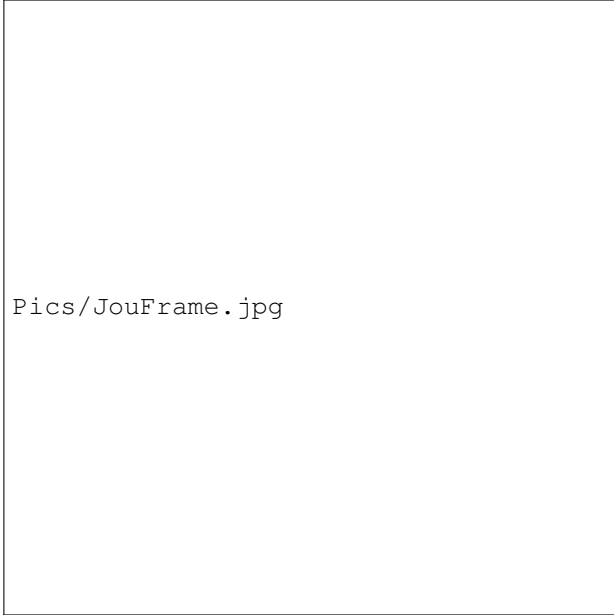


Fig. 14. Aerial frame of the Jounieh study area.

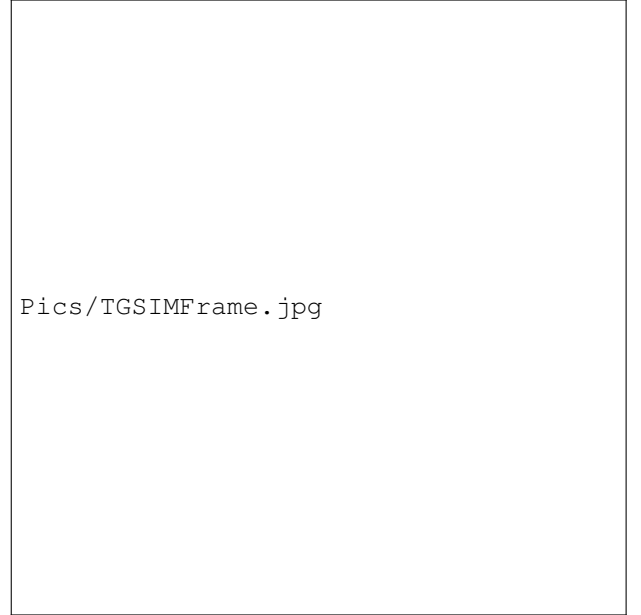


Fig. 15. Aerial frame of the TGSIM Foggy Bottom study area

B. Jounieh Roundabout

is reconstructed into two Kalman-filter movement streams ($\text{lane_kf} \in \{1, 2\}$). Rather than relying on painted lane IDs—which are only weakly respected—the spatial reference for utility is a data-driven PCA tube per $(\text{run_id}, \text{lane_kf})$: a centerline for Frenet/direction terms and lower/upper bounds for U_{path} (Fig. 16a). Class-1 passenger trajectories (190,767 valid kinematic transitions from 368 tracks; median speed 8.15 m/s) form the ego pool for the main-text calibration. This is the only environment on which the residual policy is trained and benchmarked.

A second Lebanon aerial collection covers a compact multi-arm roundabout (about $56 \text{ m} \times 31 \text{ m}$) rather than a linear corridor. Circulation wraps a large central island and a smaller splitter island; several entries and exits produce crossing and weaving at low speed (median 1.88 m/s; 85th percentile 2.70 m/s). A single roadway polygon—outer curb minus the two islands—is used for every vehicle (Fig. 16b). After the duration/speed/path filter, 479 class-2 tracks remain for calibration. Because there is no unique downstream tangent, directional and proximity terms cannot be taken from a PCA corridor; they are instead destination-based (Section F).

TABLE IX
RESIDUAL PPO TRAINER AND MAPPO BASELINE OPTIMIZATION SETTINGS.

Residual MARL (shared PPO)	Value	MAPPO baseline	Value
Actor / critic MLP	$d_{\text{obs}} \rightarrow 128 \rightarrow 128$	Actor MLP	$d_{\text{obs}} \rightarrow 128 \rightarrow 128$
Actor output	$\tanh$ -scaled $\Delta\Theta$ (9-D)	Critic input	$6N$ joint + d_{obs}
Critic input	local o_i only	Critic MLP	$d_{\text{state}} \rightarrow 128 \rightarrow 128 \rightarrow 1$
Hidden width	128	Hidden width	128
Learning rate	3×10^{-4}	Actor LR / critic LR	$3 \times 10^{-4} / 10^{-3}$
PPO clip ϵ	0.2	PPO clip ϵ	0.2
Value coef. c_v	0.5	Value coef. c_v	0.5
Entropy coef. c_H	0.01	Entropy coef. c_H	0.01
GAE λ	0.95	GAE λ	0.95
Discount γ	0.99	Discount γ	0.99
PPO epochs / update	4	PPO / critic epochs	4 / 4
Minibatch size	512	Minibatch size	256
Episodes / update	4	Episodes / update	4
Gradient clip	0.5	Gradient clip	0.5
Policy updates	100	Policy updates	80
Init. log σ (actor)	-1.2	Init. log σ (actor)	-1.6
Collision penalty (train)	8	Collision penalty (train)	8
Observation normalization	Scale	$\Delta\Theta$ tanh range	Max Δ
s_i	$/L_{\text{corridor}}$	$\Delta S_v, \Delta S_\theta$	1.5
n_i	$/8$	ΔS_d	2.0
v_i	$/16$	Δw_c	250
$\tilde{\theta}_i, \tilde{\theta}_{\text{goal},i}$	$/\pi$	$\Delta \xi_i, \Delta \gamma$	1.0
$d_{\text{lower}}, d_{\text{upper}}$	$/12$	Δw_ℓ	15
$\Delta x^{\text{body}}, \Delta y^{\text{body}}$	$/60, /12$	$\Delta \sigma_{\parallel}, \Delta \sigma_{\perp}$	1.5, 0.6
$\Delta v_x^{\text{body}}, \Delta v_y^{\text{body}}$	$/16$		
Θ clip ranges after $\Theta_{\text{base}} + \Delta\Theta$: $S_\theta, S_v, S_d \in [0.05, 12]$; $\xi_i \in [1, 10]$; $\gamma \in [0.05, 10]$; $w_c \in [0.01, 1200]$; $w_\ell \in [0.1, 300]$; $w_x, w_y \in [0.05, 15]$; β			

TABLE X
CALIBRATION-READY TRAJECTORY SAMPLES USED IN THIS APPENDIX. COUNTS REFER TO THE EGO CLASS LISTED IN THE SECOND COLUMN.
DURATIONS AND SPEEDS ARE MEDIANS OVER EGO TRACKS.

Site	Tracks	Med. duration (s)	Med. speed (m/s)	v_{85} / v_{99} (m/s)	Extent (m)	Geometry
Lebanon Highway	368	57.8	8.15	11.6 / 16.3	505×433	PCA tubes (2 rec. $\times$ 2 streams)
Jounieh roundabout	479	24.4	1.88	2.70 / 3.89	56×31	outer curb + 2 islands
TGSIM Foggy Bottom	2773	25.2	1.61	9.59 / 16.3	197×224	union street curb

C. TGSIM Foggy Bottom

The third environment is the publicly released TGSIM Foggy Bottom trajectory set [18], [19], collected with overlapping infrastructure cameras around the George Washington University campus in Washington, DC. The reconstructed street graph spans roughly $200\text{ m} \times 230\text{ m}$ of signalized and stop-controlled urban links with right-angle turns (Fig. 16c). Provided lane polygons are unioned into one curb (exterior plus interior holes) that is shared by all IDs. The prepared file retains mixed road-user types after dropping parked/short tracks; we calibrate on class-3 passenger cars (2773 tracks). Speeds are strongly bimodal: the median (1.61 m/s) reflects queues and turning, while the 85th/99th percentiles (9.59 / 16.3 m/s) capture unconstrained urban running. As at the roundabout, each track's last observed position is treated as a destination on the roadway polygon.

APPENDIX F ENVIRONMENT-BASED CALIBRATION RESULTS

We reuse the twelve-parameter search, composite objective (29), and vehicle-wise train/validation/test split of Section V (400 random trials, 3 restarts, 300/100/100 closed-loop windows of horizon $H = 8$ steps). Only the spatial encoding of U_{dir} , U_{dist} , and U_{path} changes with the site; U_{speed} and U_{col} keep the same kinematic and probabilistic forms. Collision-kernel scales ($\sigma_{\parallel}, \sigma_{\perp}$) remain free parameters.

a) *Corridor versus destination-plus-curb.*: On the free-way, $\hat{t}_i$ is the local PCA-tube tangent and U_{dist} is a Frenet distance to a short-horizon reference along that tube; U_{path} saturates outside the lower/upper PCA bounds. On the two urban sites this corridor is undefined. Each ego ID is assigned the destination $g_i = (x_i^{\text{final}}, y_i^{\text{final}})$ equal to its last observed point (utility_frame=destination). When a site polygon is supplied, heading incentive and proximity follow the on-road path toward g_i on that polygon (around islands and city

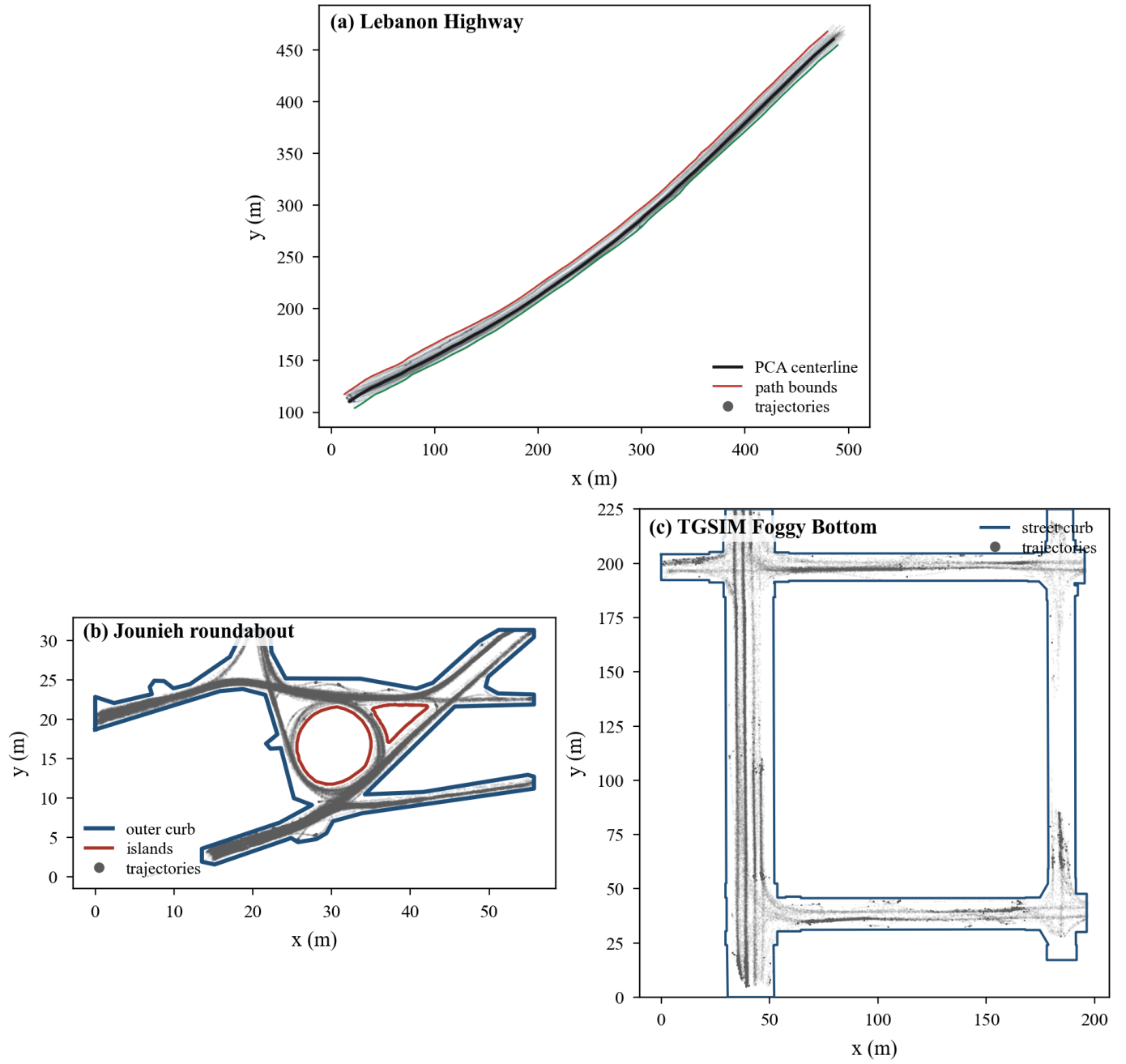


Fig. 16. Reconstructed geometry and trajectories for the three calibration environments. (a) Lebanon Highway: PCA centerline (black) and path-adherence bounds (red/green) for one recording and one direction ($run_id=2$, $lane_kf=1$). (b) Jounieh roundabout: outer curb (blue) and interior islands (red). (c) TGSIM Foggy Bottom: union street curb (blue), clipped and closed at $y = 225$ m.

blocks) rather than the Euclidean chord, and path cost grows with off-road distance. Desired speed remains the track-level desired speed used in the freeway fit; urban kinematic limits follow the site speed distribution. The Jounieh and TGSIM JSON files used here record that polygon path; they are not a second corridor fit.

b) Closed-loop scores.: Held-out tracking is strongest on the freeway (test $L_{cl} = 0.565$), where a unique downstream tangent exists and the PCA tube already concentrates the data.

The roundabout remains usable (0.714) despite low speeds and circulating geometry. Foggy Bottom is the most demanding (0.992): vehicles approach a destination through intersections, queues, and right-angle turns, so an eight-step utility rollout has more ways to miss the observed branch. In all three cases the validation-selected working point generalizes (test is within about 5% of validation), so the gap across sites is geometric rather than an overfit of the search.

TABLE XI

ENVIRONMENT-BASED UTILITY CALIBRATION. TEST L_{cl} IS CLOSED-LOOP TRACKING LOSS ON HELD-OUT VEHICLES ($H = 8$). RANK IS THE MEAN ONE-STEP RANK OF THE OBSERVED-LIKE CANDIDATE DURING SEARCH. WORKING (w_ℓ, w_c) COME FROM THE VALIDATION-SELECTED VECTOR (ROBUST MEDIAN ON THE FREEWAY; BEST-ON-VALIDATION AT THE URBAN SITES). ALL THREE FITS ARE WEAKLY IDENTIFIED.

Site	Frame	Test L_{cl}	Val L_{cl}	Rank	w_ℓ	w_c	Near-opt. trials
Lebanon Highway	corridor	0.565	0.576	5.72	77.6	300	3
Jounieh roundabout	destination + curb	0.714	0.733	37.9	9.17	71.4	3
TGSIM Foggy Bottom	destination + curb	0.992	0.965	27.8	68.4	5.92	10 [†]

[†]Top-ten fallback: fewer than three scored trials met the 5% near-optimal cutoff.

TABLE XII

WORKING UTILITY PARAMETERS AT EACH SITE. FREEWAY VALUES ARE THE ROBUST MEDIAN OF THE NEAR-OPTIMAL CLOUD SELECTED ON VALIDATION WINDOWS; URBAN VALUES ARE THE BEST TRIAL ON VALIDATION WINDOWS.

Parameter	Highway	Jounieh	TGSIM
S_θ	4.13	3.88	8.46
S_v	2.51	3.89	9.68
ξ_i	4.04	9.83	4.02
S_d	9.43	8.44	7.76
γ	0.56	3.63	2.57
w_x	4.85	5.11	9.49
w_y	5.83	4.81	2.20
w_c	300	71.4	5.92
w_ℓ	77.6	9.17	68.4
β	2.14	6.39	4.59
$\sigma_{\parallel}$ (m)	0.89	2.23	0.67
$\sigma_{\perp}$ (m)	0.59	1.42	2.46

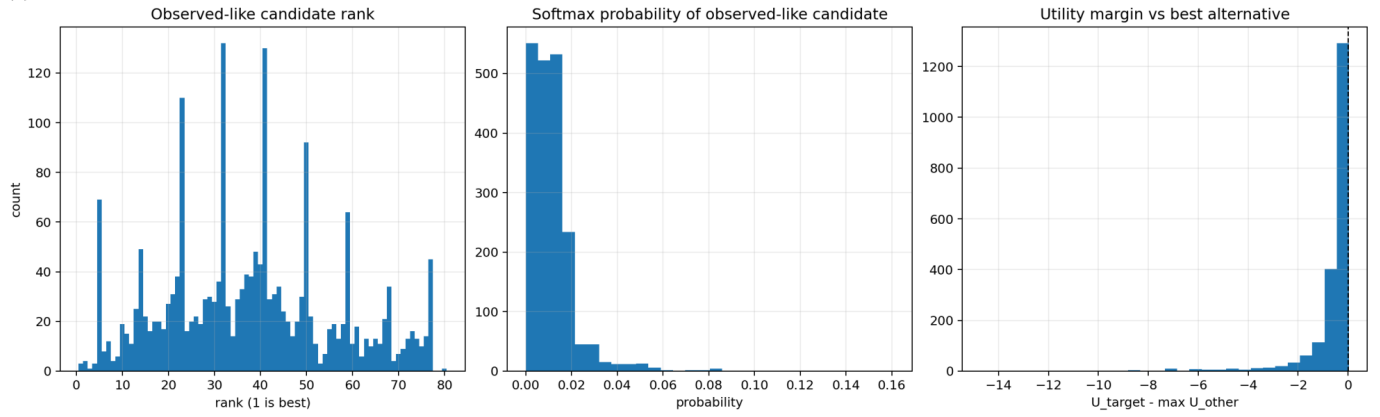
more heavily than the freeway, as would be expected when progress is defined by reaching a destination through a sparse street graph rather than staying in a tube.

e) Scope.: These results support transferring the five-term prior across environments *provided* the spatial frame is rewritten for the site (corridor tubes versus destination on a curb polygon). They do not show that one numerical θ is universal, and they do not replace the freeway residual-MARL study: urban closed-loop errors are larger, identifiability is weaker, and no residual policy was trained at the roundabout or at Foggy Bottom. Per-track overlays remain mixed—many vehicles stay inside the curb and trend toward their destination, while a subset still freeze, cut a corner, or take the wrong exit arm—which is why the population test loss, rather than any single ID, is the quantity reported in Table XI.

c) One-step ranking and identifiability.: One-step ranking of the observed-like candidate remains informative on the freeway (mean rank 5.72) but is close to chance-competitive on both urban sites (Fig. 17): mean ranks 37.9 (Jounieh) and 27.8 (TGSIM) on the discrete acceleration–steering grid, with softmax mass on the target action typically below 0.05. Soft margins $U_{\text{target}} - \max U_{\text{other}}$ nevertheless concentrate near zero, so the observed action is often a near-maximizer even when it is not rank one. This is consistent with the weakly identified verdict at every site (three near-optimal trials on the freeway and roundabout; top-ten fallback at Foggy Bottom). As in the main text, we therefore treat the working vector as a region of acceptable short-horizon behavior, not a unique structural estimate.

d) What the weights say about geometry.: Table XII shows that the same parameter box produces quite different operating points. Path weight w_ℓ is large on the freeway (77.6), where the PCA tube is a tight behavioral constraint, and remains large at Foggy Bottom (68.4), where the curb is the only thing preventing an intersection-cutting chord to the destination. At the roundabout w_ℓ collapses to 9.17: once U_{dir} and U_{dist} already follow the circulating on-road path around the islands, an additional curb penalty is only weakly identified and several near-optimal trials trade it off against S_θ . Collision weight w_c is likewise site-dependent (300 / 71 / 6), matching the main-text observation that w_c is poorly constrained by short windows in which collisions are rare. Foggy Bottom loads directional and speed terms ($S_\theta = 8.46$, $S_v = 9.68$)

(a) Jounieh roundabout



(b) TGSIM Foggy Bottom

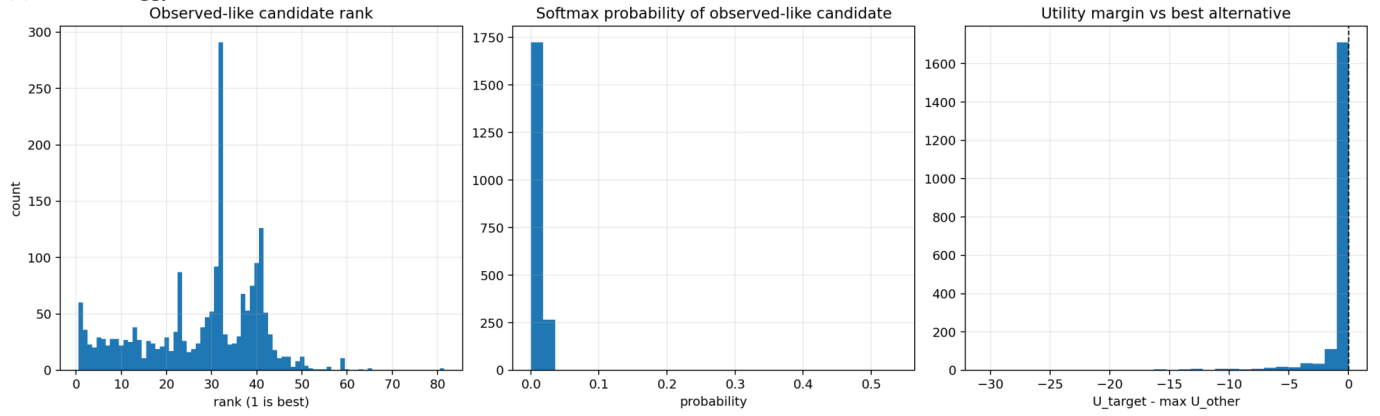


Fig. 17. Urban one-step diagnostics under the destination-plus-curb encoding. Left to right in each row: rank of the observed-like candidate, softmax probability of that candidate, and utility margin relative to the best alternative. (a) Jounieh roundabout. (b) TGSIM Foggy Bottom. Rank histograms are much flatter than the freeway fit in Fig. 1, while margins still pile up near zero.